
Parameter Decorrelation via Transition-Variance Alignment for Multivariate Time-series Forecasting

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Abstract

Multivariate time-series forecasting (MTSF) learns from high-dimensional covariates with strong temporal dependence, periodic structure, and cross-variable correlations. While modern pipelines often mitigate non-stationarity through instance-wise normalization and decomposition, these interventions operate at the data level and do not directly control dependence that can emerge among the parameters during training. We study MTSF optimization from a parameter-decorrelation viewpoint. Modeling stochastic optimization as a Markov chain in parameter space and leveraging its stochastic differential equation interpretation, we use the per-step transition-variance induced by gradient noise as a tractable signal for optimization-induced dependence and update uncertainty. This signal can empirically inflate during training; we theoretically show that such inflation can degrade generalization diagnostics. Motivated by this mechanism, we propose transition-variance alignment (TVA), an architecture-agnostic procedure that regulates transition-variance by smoothly gating the step size based on the mismatch between an estimated noise scale and a chosen target. TVA maintains effective transition-variance near a prescribed scale without architectural changes, incurs negligible overhead, and integrates seamlessly with diverse methods. Across real-world multivariate benchmarks, TVA consistently improves forecasting accuracy.

1. Introduction

Time-series forecasting (TSF) aims to predict future values from historical observations and supports planning and risk mitigation in domains such as energy systems, traffic management, and weather prediction (Quinlan, 1986; Cortes & Vapnik, 1995; Vapnik et al., 1996; Friedman, 2001; Barlin et al., 2013; Chen & Guestrin, 2016). In multivariate time-series forecasting (MTSF), the forecaster additionally exploits covariate signals to improve predictive performance, yielding a practically important but statistically more demanding setting (Wen et al., 2023; Qiu et al., 2024; Luo & Wang, 2024). The central challenge is to learn temporal dynamics in a high-dimensional covariate space that is both cross-correlated and strongly temporally dependent.

Deep neural networks (DNNs) have become a dominant solution paradigm for MTSF, spanning architectures from linear and multilayer perceptron (MLP) models to Transformer-based forecasters for long-horizon prediction (Zeng et al., 2023; Zhang & Yan, 2023; Nie et al., 2023; Lin et al., 2024a; Cai et al., 2024; Liu et al., 2024). Despite strong empirical progress, time-series data commonly exhibit pronounced periodicity and inter-sample dependence; consequently, training examples obtained from sliding windows are far from independent and identically distributed (i.i.d.) (Lin et al., 2024a). This mismatch can distort optimization dynamics and exacerbate generalization issues, particularly in long-horizon settings where repetitive structure and non-stationary shifts coexist.

A substantial body of work addresses these challenges via data decorrelation. Instance-wise normalization methods (Kim et al., 2022; Fan et al., 2023; Liu et al., 2023; Ye et al., 2024) normalize each sample using its own statistics, attenuating nuisance components such as sample-specific bias and scale before restoring them at the output. Complementary, decomposition-based models (Zeng et al., 2023; Yu et al., 2024; Lin et al., 2024a) separate trend/seasonality or periodic components from residuals to isolate predictable structure. These methods operate primarily at the level of representation and have proven effective for mitigating distribution shift and periodic confounds. However, they do not explicitly regulate correlations that may emerge among parameters when the model is repeatedly updated on tempo-

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rally dependent data.

In this work, we highlight the risk of over-correlated parameters in MTSF training and propose a mechanism to diagnose and mitigate it. We view stochastic optimization as inducing a discrete-time Markov chain on parameters and approximate its dynamics via stochastic differential equations (SDEs) (Li et al., 2019; 2021; Malladi et al., 2022). Under this perspective, each training step defines a transition kernel whose variance captures the uncertainty injected by gradient noise at that step. We use this transition-variance as a practical signal of optimization-induced dependence and update uncertainty. Empirically, we observe that the signal can inflate rather than stabilize; analytically, we show that such inflation can enlarge PAC-Bayes (Seeger, 2002) complexity terms, providing a mechanistic diagnostic for reduced generalization.

Guided by this insight, we propose transition-variance alignment (TVA), a simple architecture-agnostic method for controlling transition-variance during training. At each iteration, TVA estimates the transition-variance for the gradient-noise scale and compares it to a target level. TVA then smoothly gates the step size using a log-ratio discrepancy rule, shrinking the effective step size when the estimated noise scale deviates from the target and leaving it near its baseline when the scales are well matched. This adaptive regulation keeps the effective transition-variance close to the target, preventing runaway growth of the signal and stabilizing the induced parameter distribution. TVA requires no changes to the forecasting backbone and introduces only lightweight computations, making it compatible with linear, MLP, and Transformer-based forecasters. We validate TVA across diverse MTSF backbones, showing consistent improvements in training stability and forecasting accuracy on real-world benchmarks.

2. Related Works

2.1. Time-series Forecasting Models

TSF has long been studied through statistical modeling and has increasingly incorporated machine learning techniques to capture nonlinear temporal dependencies and complex covariate effects (Quinlan, 1986; Cortes & Vapnik, 1995; Vapnik et al., 1996; Friedman, 2001; Barlin et al., 2013; Chen & Guestrin, 2016). As deployments have grown in scope and dimensionality, research emphasis has shifted toward MTSF, which leverages multiple covariates to improve predictive accuracy and robustness (Wen et al., 2023; Qiu et al., 2024; Luo & Wang, 2024). Contemporary MTSF methods span a broad architectural spectrum, ranging from linear and MLP-based models to Transformer-style forecasters designed for long-horizon prediction (Zeng et al., 2023; Nie et al., 2023; Zhang & Yan, 2023; Lin et al., 2024a;

Cai et al., 2024; Liu et al., 2024). A recurring theme is the trade-off between architectural complexity and inductive bias, with several works reporting that carefully designed lightweight models can be highly competitive, while others show that attention-based mechanisms can be effective when paired with suitable tokenization schemes, channel modeling choices, or structural priors (Nie et al., 2023; Das et al., 2023; Lin et al., 2024a). Our work highlights a complementary dimension by showing how temporally dependent sampling, and the resulting optimization dynamics, shape the learned parameter distribution.

2.2. Data Decorrelation

A prominent strategy for improving forecasting under non-stationarity is to reduce redundant or unstable components in the input through normalization. Instance-wise normalization methods compute sample-specific statistics and normalize each window independently, suppressing nuisance factors such as shifting level and scale (Ogasawara et al., 2010; Passalis et al., 2019; Kim et al., 2022; Fan et al., 2023; Liu et al., 2023). Reversible normalization frameworks introduce a normalize–forecast–denormalize pipeline that removes window statistics prior to prediction and reinserts them at reconstruction time, improving robustness to distribution shift (Kim et al., 2022). More recent variants operate at a finer temporal granularity to better accommodate within-window non-stationarity (Fan et al., 2023; Liu et al., 2023). Frequency-aware normalization has also been proposed to address evolving seasonal components by operating in the frequency domain (Ye et al., 2024). Another widely used family of approaches performs seasonal–trend decomposition (STD), decomposing each input window into components such as trend and seasonality that are modeled separately (Wu et al., 2021; Zhou et al., 2022; Fan et al., 2022; Zeng et al., 2023). Many STD-based designs use moving-average filters to extract slowly varying trends, while more flexible approaches replace fixed filters with learnable decomposition kernels to adapt to dataset-specific dynamics (Yu et al., 2024; Lin et al., 2024b). By isolating periodic structure from residual variation, decomposition can stabilize learning and improve extrapolation. Normalization and decomposition primarily target data correlation and non-stationarity. Our approach is complementary in that we focus on correlations that can emerge in parameter dynamics during training when mini-batches are temporally dependent.

3. Methodology

In this section, we develop the parameter-decorrelation perspective and study how temporally dependent sampling influences training and the induced parameter distribution. Specifically, we formalize MTSF and stochastic optimiza-

tion (Section 3.1), derive a Gaussian transition model from an SDE viewpoint and introduce a scalar transition-variance signal (Section 3.2), and use a PAC-Bayes bound as a diagnostic to show how signal inflation can render complexity terms vacuous (Section 3.3). Next, we justify transition-wise KL control as a sufficient surrogate for controlling marginal KL drift along the induced parameter distribution (Section 3.4). Finally, we derive the TVA rule (Section 3.5), together with a lightweight recursion and a practical training objective (Section 3.6). Proofs of the lemmas and the theorem are provided in Appendix A.1 through Appendix A.3, and pseudo-code is provided in Appendix A.4.

3.1. Problem Settings

We consider MTSF, where each training example consists of a look-back window of covariates $\mathbf{x} \in \mathcal{X}$ and a look-forward horizon $\mathbf{y} \in \mathcal{Y}$ to be predicted. Let $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ be a forecaster parameterized by $\hat{\theta} \sim \theta \in \mathbb{R}^d$, inducing a conditional model $p(\mathbf{y}|\mathbf{x}; \theta)$. We write $z = (\mathbf{x}, \mathbf{y}) \in \mathcal{Z}$ and let $S = \{z_i\}_{i=1}^n$ denote a mini-batch of size n . The population risk is

$$R(\theta) \triangleq \mathbb{E}_{z \sim D} [\ell(\theta, z)], \quad (1)$$

where $\ell(\theta, z) = -\log p(\mathbf{y}|\mathbf{x}; \theta)$. In practice, we minimize the empirical risk

$$L_S(\theta) \triangleq \frac{1}{n} \sum_{i=1}^n \ell(\theta, z_i) \quad (2)$$

via a stochastic optimizer. Abstractly, one update can be written as

$$\hat{\theta}_k = \hat{\theta}_{k-1} + \Delta\theta_k, \Delta\theta_k = -\eta_k \hat{\nabla} L_{S_k}(\hat{\theta}_{k-1}), \quad (3)$$

where S_k is the time-series mini-batch used at step k and η_k is the step size.

In MTSF, sliding-window sampling induces strong dependence among mini-batches, so the stochastic gradients $\hat{\nabla} L_{S_k}$ can be temporally correlated across k . This temporal dependence can propagate into the parameter trajectory $\{\hat{\theta}_k\}$, reducing the effective mixing of the induced parameter distribution and increasing the risk of overfitting. Our method addresses this phenomenon by controlling parameter transitions between consecutive optimization steps.

3.2. Probabilistic Framework: Gaussian Transition Model and Signal for Temporal Correlation

In a small-step regime, stochastic-gradient methods admit SDE approximations that make explicit the role of gradient noise (Li et al., 2019; 2021; Malladi et al., 2022). A standard Itô form is

$$d\theta_t = -g(\theta_t) dt + \sqrt{\eta} B(\theta_t) dW_t, \Sigma(\theta_t) = B(\theta_t)B(\theta_t)^\top, \quad (4)$$

where $g(\theta)$ denotes a empirical gradient field, W_t is a standard Brownian motion, and $\Sigma(\theta_t)$ is the local covariance of the stochastic gradient noise. Specifically, we define: $\Sigma(\theta_t) = 1/t \sum_{\tau=1}^t (g(\theta_\tau) - \bar{g}(\theta_\tau))(g(\theta_\tau) - \bar{g}(\theta_\tau))^\top$ with $\bar{g}(\theta_t) = 1/t \sum_{\tau=1}^t g(\theta_\tau)$ as the mean gradient.

The density $p(\theta, t)$ then evolves according to the Fokker-Planck-Kolmogorov equation (Särkkä & Solin, 2019)

$$\partial_t p(\theta, t) = \sum_{i=1}^d \frac{\partial}{\partial \theta^i} (p(\theta, t) [g(\theta)]_i) \quad (5)$$

$$+ \frac{\eta}{2} \sum_{i=1}^d \sum_{j=1}^d \frac{\partial^2}{\partial \theta^i \partial \theta^j} (p(\theta, t) [\Sigma(\theta)]_{ij}). \quad (6)$$

Since solving Eq. (6) is intractable for modern deep networks, we adopt a Gaussian assumed-density approximation $p(\theta, t) \approx \mathcal{N}(\theta|m_t, V_t)$ (Ansari et al., 2023). Linearizing the drift around m_t gives $g(\theta) \approx g(m_t) + G_t(\theta - m_t)$ with Jacobian $G_t = \nabla_\theta g(m_t)$. The corresponding mean and covariance dynamics satisfy

$$\frac{dm_t}{dt} = -g(m_t), \frac{dV_t}{dt} = -G_t V_t - V_t G_t^\top + \eta \Sigma(m_t), \quad (7)$$

which is a standard consequence of Gaussian approximations for SDEs.

Discrete-time transition kernel. Our algorithm operates per optimizer step, so we adopt a discrete-time transition model. Decompose the stochastic gradient into its conditional mean plus noise:

$$\hat{\nabla} L_{S_k}(\theta) = g_k(\theta) + \xi_k, \mathbb{E}[\xi_k|\theta] = 0, \text{Cov}(\xi_k|\theta) = \Sigma_k(\theta), \quad (8)$$

where $g_k(\theta)$ is the conditional mean gradient under the mini-batch sampling mechanism at step k , and $\Sigma_k(\theta)$ is the conditional gradient-noise covariance. Substituting Eq. (8) into the update rule Eq. (3) yields the Gaussian transition approximation, conditional on $\hat{\theta}_{k-1}$,

$$\hat{\theta}_k | \hat{\theta}_{k-1} \approx \mathcal{N}(\hat{\theta}_{k-1} - \eta_k g_k(\hat{\theta}_{k-1}), \eta_k^2 \Sigma_k(\hat{\theta}_{k-1})). \quad (9)$$

Eq. (9) makes explicit that the effective step size η_k controls the transition covariance quadratically, which provides a direct lever for regulating the uncertainty of parameter moves across steps.

Transition-variance signal. Estimating the full matrix $\Sigma_k(\hat{\theta}_{k-1})$ is typically infeasible, so we track its mean-field scale $\sigma_k^2 \triangleq \frac{1}{d} \text{tr}(\Sigma_k(\hat{\theta}_{k-1}))$, and approximate $\Sigma_k \approx \sigma_k^2 I$. With this approximation, Eq. (9) simplifies to

$$p(\theta_k | \theta_{k-1}) = \mathcal{N}(\theta_k | \theta_{k-1} - \eta_k g_k(\theta_{k-1}), \sigma_k^2 \eta_k^2 I). \quad (10)$$

where the scalar $\sigma_k^2 \eta_k^2$ serves as our transition-variance signal. This quantity measures the scale of step-to-step update

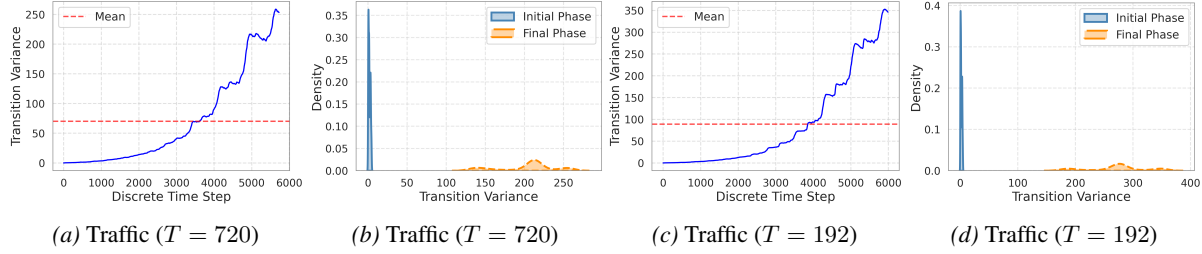


Figure 1. Transition-variance dynamics and their distribution in the Traffic dataset. Subfigures (a) and (c) illustrate the variance dynamics over discrete time steps, while (b) and (d) present kernel density estimation plots of the transition-variance computed from the first 20% (Initial Phase) and last 20% (Final Phase) segments. The look-back window L is fixed to 96, with prediction lengths T of 720 for (a, b) and 192 for (c, d). In both long- and short-horizon forecasting settings, the transition-variance increases progressively, indicating non-stationary behavior. Additional results on various datasets are presented in Appendix C.1.

uncertainty; in MTSF, temporally persistent gradient-noise effects induced by overlapping windows can cause this signal to vary non-trivially over training, thereby providing a practical handle on optimization-induced dependence in the parameter trajectory.

3.3. Diverging Transition-Variance and a PAC-Bayes Diagnostic

Empirically (see Fig. 1), the transition-variance signal increase over training time step in both long-horizon and shorter-horizon TSF tasks. Persistent inflation of this quantity suggests that the induced parameter distribution spreads over time and drifts away from any fixed reference distribution. As a consequence, PAC-Bayes generalization certificates—whose complexity term depends on a KL divergence between a data-dependent posterior and a fixed prior—can become vacuous.

We adopt the PAC-Bayes inequality stated in a convenient λ -form for bounded losses (Seeger, 2002; Alquier, 2021; Rothfuss et al., 2023). Let ρ be a data-dependent posterior distribution over parameters and π a fixed prior independent of the sample S . Assume $\ell(\theta, z) \in [0, C]$ for all (θ, z) .

Lemma 3.1 (PAC-Bayes certificate worsens under diverging posterior variance). *Fix $\delta \in (0, 1)$ and any $\lambda > 0$ chosen independently of S . With probability at least $1 - \delta$ over $S \sim D^n$, simultaneously for all distributions ρ on $\mathbb{R}^d$,*

$$\mathbb{E}_{\theta \sim \rho} [R(\theta)] \leq \mathbb{E}_{\theta \sim \rho} [L_S(\theta)] \quad (11)$$

$$+ \frac{\lambda C^2}{8n} + \frac{\text{KL}(\rho \parallel \pi) + \log(1/\delta)}{\lambda}. \quad (12)$$

Assume $\pi = \mathcal{N}(0, \sigma^2 I)$ and $\rho_k = \mathcal{N}(m_k, s_k^2 I)$, where $(s_k^2)_{k \geq 1}$ is nondecreasing and diverges ($s_k^2 \rightarrow \infty$) while $(\|m_k\|_2)_{k \geq 1}$ is bounded. Then, for any fixed $\lambda > 0$, the PAC-Bayes complexity term

$$\mathcal{C}_k(\lambda) \triangleq \frac{\text{KL}(\rho_k \parallel \pi) + \log(1/\delta)}{\lambda}$$

is eventually nondecreasing and satisfies $\mathcal{C}_k(\lambda) \rightarrow \infty$ as $k \rightarrow \infty$. Consequently, the bound (12) worsens for all

sufficiently large k and becomes vacuous along any such variance-diverging trajectory.

Lemma 3.1 motivates stabilizing the variance of the induced parameter distribution. In our framework, this is achieved by regulating the transition-variance appearing in Eq. (10).

3.4. Parameter Decorrelation via Transition-KL Control

We focus on the PAC-Bayes complexity term, which depends on a marginal KL divergence (e.g., $\text{KL}(\rho_k \parallel \pi)$). Our key move is to control an upper bound on a marginal KL by a sum of transition KL divergences between Markov kernels. This motivates a local control strategy in which keeping each step-to-step transition close to a reference transition prevents the induced marginal distribution from drifting too far from the corresponding reference marginal.

Let $(P_k)_{k \geq 1}$ and $(Q_k)_{k \geq 1}$ be Markov kernels on $\mathbb{R}^d$. Starting from initial distributions p_0 and q_0 , define the induced marginals recursively by

$$p_k(\cdot) = \int P_k(\cdot | \theta) p_{k-1}(d\theta), q_k(\cdot) = \int Q_k(\cdot | \theta) q_{k-1}(d\theta).$$

Lemma 3.2 (Transition KL control implies marginal KL control). *For all $k \geq 1$,*

$$\begin{aligned} \text{KL}(p_k \parallel q_k) &\leq \text{KL}(p_0 \parallel q_0) \\ &+ \sum_{i=0}^{k-1} \mathbb{E}_{\theta_i \sim p_i} [\text{KL}(P_{i+1}(\cdot | \theta_i) \parallel Q_{i+1}(\cdot | \theta_i))]. \end{aligned} \quad (13)$$

In particular, if $p_0 = q_0$, then reducing the expected transition KL terms tightens an upper bound on $\text{KL}(p_k \parallel q_k)$.

Lemma 3.2 justifies a local control strategy by minimizing the expected transition KL between a data-driven kernel P_k and a reference kernel Q_k ; we control an upper bound on the marginal KL drift along the induced parameter distribution.

3.5. Transition-Variance Alignment (TVA)

Motivated by the isotropic transition kernel in Eq. (10), we define a data-driven transition kernel

$$P_k(\cdot|\theta_{k-1}) = \mathcal{N}\left(\cdot \mid \theta_{k-1} - \eta_k g_k(\theta_{k-1}), \sigma_k^2 \eta_k^2 I\right), \quad (14)$$

and a reference kernel with a target transition-variance $\sigma_\lambda^2 \eta^2$:

$$Q_k(\cdot|\theta_{k-1}) = \mathcal{N}\left(\cdot \mid \theta_{k-1} - \eta_k g_k(\theta_{k-1}), \sigma_\lambda^2 \eta^2 I\right). \quad (15)$$

We match the drift so that the transition KL depends only on covariance mismatch, yielding a stable control signal. For Gaussians with matched means and isotropic covariances, letting $s_k^2 = \sigma_k^2 \eta_k^2$ and $u^2 = \sigma_\lambda^2 \eta^2$, the transition KL admits the closed form

$$\text{KL}(P_k\|Q_k) = \frac{d}{2} \left(\frac{s_k^2}{u^2} - 1 - \log \frac{s_k^2}{u^2} \right). \quad (16)$$

Controlling the transition-variance via a step-size gate.

We parameterize the step size as

$$\eta_k = \alpha_k \eta, \quad (17)$$

where $\alpha_k \in (0, 1]$ and η is a baseline step size. The data-driven transition-variance becomes $s_k^2 = \sigma_k^2 \alpha_k^2 \eta^2$. To prevent variance blow-up while keeping the update smooth, we use the symmetric log-ratio discrepancy rule

$$\begin{aligned} \alpha_k &\triangleq \exp\left(-\left|\log(\sigma_k + \varepsilon) - \log \sigma_\lambda\right|\right) \\ &= \min\left\{\frac{\sigma_k + \varepsilon}{\sigma_\lambda}, \frac{\sigma_\lambda}{\sigma_k + \varepsilon}\right\}, \end{aligned} \quad (18)$$

with a small $\varepsilon > 0$ for numerical stability. With Eq. (17)–(18), the controlled transition becomes

$$p(\theta_k|\theta_{k-1}) = \mathcal{N}\left(\theta_k \mid \theta_{k-1} - \alpha_k \eta g_k(\theta_{k-1}), \sigma_k^2 (\alpha_k \eta)^2 I\right). \quad (19)$$

This gate has three useful properties. First, $\alpha_k \leq 1$ always and $\alpha_k \approx 1$ when $\sigma_k \approx \sigma_\lambda$. Second, in the early phase where σ_k is small relative to σ_λ , the rule behaves as $\alpha_k \approx (\sigma_k + \varepsilon)/\sigma_\lambda$, which increases the effective transition-variance smoothly as training proceeds. Third, as the mismatch grows on the log scale, α_k decreases smoothly; since $\alpha_k = \exp(-|\Delta_k|)$ with $\Delta_k = \log(\sigma_k + \varepsilon) - \log \sigma_\lambda$, the mapping $\Delta \mapsto \exp(-|\Delta|)$ is globally Lipschitz, which helps avoid abrupt changes in the controlled transition-variance and yields stable training dynamics.

The following theorem formalizes that the gate in Eq. (18) prevents variance-driven blow-up by uniformly bounding the per-step transition KL when σ_k becomes large.

Theorem 3.3 (Complexity control via adaptive transition-variance alignment). *Fix $\delta \in (0, 1)$ and assume the bounded-loss setting of Lemma 3.1. Let $\{\theta_k\}_{k \geq 0}$ be the Markov chain induced by $P_k(\cdot|\theta_{k-1})$ in Eq. (14) with gated step size $\eta_k = \alpha_k \eta$ and α_k given by Eq. (18). Let $\{q_k\}_{k \geq 0}$ be a reference chain induced by $Q_k(\cdot|\theta_{k-1})$ in Eq. (15), and denote their marginals by p_k and q_k . Assume $p_0 = q_0$. Then for every $k \geq 1$,*

$$\text{KL}(p_k\|q_k) \leq \sum_{i=0}^{k-1} \mathbb{E}_{\theta_i \sim p_i} [\text{KL}(P_{i+1}(\cdot|\theta_i)\|Q_{i+1}(\cdot|\theta_i))]. \quad (20)$$

Moreover, since P_i and Q_i have matched means and isotropic covariances,

$$\text{KL}(P_i\|Q_i) = \frac{d}{2} (r_i - 1 - \log r_i), \quad (21)$$

where $r_i \triangleq \frac{\sigma_i^2 \alpha_i^2}{\sigma_\lambda^2}$. Under the gate Eq. (18), we always have $\sigma_i \alpha_i \leq \sigma_\lambda$ and hence $r_i \leq 1$. In particular, in the variance-inflation regime $\sigma_i \geq \sigma_\lambda$, the gate yields $\alpha_i = \sigma_\lambda/(\sigma_i + \varepsilon)$ and therefore

$$r_i = \left(\frac{\sigma_i}{\sigma_i + \varepsilon}\right)^2 \in [(\frac{\sigma_\lambda}{\sigma_\lambda + \varepsilon})^2, 1), \quad (22)$$

so $\text{KL}(P_i\|Q_i)$ is uniformly bounded over this regime. Consequently, TVA prevents variance-driven explosion of the transition-KL contributions when σ_i becomes large, and thus controls an upper bound on marginal KL drift via Eq. (20).

3.6. Recursion and Training Objective

The controlled transition Eq. (19) enables a recursive approximation to the marginal parameter distribution. Let $p(\theta_k) = \mathcal{N}(\theta_k|m_k, V_k)$ denote the marginal at step k . By the Chapman–Kolmogorov equation (Särkkä & Svensson, 2023),

$$p(\theta_k) = \int p(\theta_k|\theta_{k-1}) p(\theta_{k-1}) d\theta_{k-1}, \quad (23)$$

and noting that both the marginal and transition are Gaussian under our approximation, we obtain the recursion

$$m_k = m_{k-1} - \alpha_k \eta g_k(m_{k-1}), V_k = V_{k-1} + \sigma_k^2 (\alpha_k \eta)^2 I, \quad (24)$$

with $m_0 = \hat{\theta}_0$ and $V_0 = 0$. This representation separates mean dynamics from covariance dynamics, and makes explicit how the TVA gate α_k regulates the per-step increase in marginal variance.

Prediction and loss. Exact Bayesian prediction would average the predictive distribution over $p(\theta_k)$:

$$p(\mathbf{y}|\mathbf{x}; m_k) \approx \int p(\mathbf{y}|\mathbf{x}, \theta) p(\theta_k) (d\theta). \quad (25)$$

Table 1. Multivariate time-series forecasting results with prediction length $T \in \{96, 192, 336, 720\}$. Results are averaged over all prediction lengths. ECL and WTH denote the Electricity and Weather datasets, respectively. The best and second-best results are highlighted in **bold** and underlined, respectively.

Model	MLP		MLP+TVA		iTransformer		PatchTST		MSGNet		Crossformer		TimesNet	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.448	0.435	0.438	0.432	0.449	0.442	0.451	0.439	0.469	0.465	0.580	0.554	0.468	0.458
ETTh2	<u>0.374</u>	<u>0.400</u>	0.371	0.397	0.379	0.406	0.395	0.412	0.393	0.418	0.805	0.642	0.406	0.423
ETTm1	0.393	0.399	0.389	<u>0.398</u>	0.413	0.413	<u>0.390</u>	0.397	0.425	0.422	0.502	0.511	0.409	0.416
ETTm2	0.290	0.335	0.278	0.324	0.293	0.337	<u>0.288</u>	0.334	0.291	<u>0.331</u>	0.482	0.487	0.296	0.333
ECL	0.218	0.299	<u>0.195</u>	<u>0.285</u>	0.190	0.280	<u>0.259</u>	0.346	0.337	0.410	0.327	0.398	0.238	0.331
Solar	0.277	0.352	0.234	0.310	0.298	0.315	0.349	0.369	0.625	0.521	<u>0.272</u>	<u>0.311</u>	0.316	0.316
Traffic	0.508	0.327	<u>0.452</u>	<u>0.300</u>	0.435	0.289	0.554	0.351	0.656	0.355	0.598	0.329	0.645	0.344
WTH	0.273	0.291	0.260	0.280	0.275	0.293	0.280	0.297	<u>0.265</u>	<u>0.290</u>	0.384	0.431	0.267	<u>0.290</u>

which is generally intractable for modern forecasters. For efficiency, we adopt a standard plug-in approximation,

$$\ell(m_k, (\mathbf{x}, \mathbf{y})) = -\log p(\mathbf{y}|\mathbf{x}; m_k). \quad (26)$$

Training objective with TVA. At iteration k , TVA computes the gate α_k via Eq. (18) and performs the gated update Eq. (3) with $\eta_k = \alpha_k \eta$. The method therefore modifies the optimization trajectory through α_k while directly regulating the transition-variance $\sigma_k^2(\alpha_k \eta)^2$ in Eq. (19). Following Theorem 3.3, this yields a principled mechanism to prevent variance-driven blow-up of the transition-KL surrogate and to stabilize the induced parameter distribution over training.

4. Experiments

In this section, we investigated the effectiveness and efficiency of TVA on standard MTSF benchmarks, comparing against widely used architectures and existing techniques. Additional analyses—including the evolution of the transition-variance under TVA, visualizations of forecasts, robustness and reproducibility, and full results for all tables—are provided in Appendix C.

4.1. Settings

Dataset and Metrics. We used eight widely adopted time-series forecasting datasets—the ETT series (ETTh1, ETTh2, ETTm1, ETTm2) (Zhou et al., 2021), Weather, Traffic, Electricity, and Solar (Wu et al., 2021)—to evaluate the proposed method. A detailed description of each dataset is provided in Appendix B.1. Following prior studies (Zhou et al., 2021; Wu et al., 2021; Zhou et al., 2022), we reported mean squared error (MSE) and mean absolute error (MAE). MSE penalizes larger deviations more heavily, while MAE measures average absolute prediction error; both metrics are standard in TSF benchmarks for their interpretability and widespread use.

Baselines. To assess forecasting accuracy relative to established methods, we compared against representative

Transformer-based forecasters, including iTransformer (Liu et al., 2024), PatchTST (Nie et al., 2023), MSGNet (Cai et al., 2024), and Crossformer (Zhang & Yan, 2023), as well as the TCN-based TimesNet (Wu et al., 2023). We additionally included DLinear (Zeng et al., 2023), CLinear (Lin et al., 2024a), LDLinear (Yu et al., 2024) as a strong and widely used linear baseline.

Implementation Details. The proposed TVA method is architecture-agnostic and can be integrated into diverse forecasting backbones by modifying only the optimizer step size according to the gate. Unless otherwise stated, we used a lightweight MLP as the primary backbone for controlled evaluations. The architecture follows the MLP design in (Lin et al., 2024a): a channel-wise input projection layer, two feedforward layers with Gaussian error linear unit (GELU) activations (Hendrycks & Gimpel, 2016), and an output layer mapping hidden representations to the prediction horizon. Additional architectural details are provided in Appendix B.2. All models were evaluated under the same experimental protocol across datasets, with prediction lengths $T \in \{96, 192, 336, 720\}$ and a fixed look-back window $L = 96$. We followed common preprocessing procedures used in prior works (e.g., iTransformer (Liu et al., 2024), PatchTST (Nie et al., 2023)), including dataset splitting and normalization. Models were trained using Adam (Kingma & Ba, 2015), and the random seed was fixed to 2026 to ensure reproducibility. The TVA target scale was selected by a greedy search over $\sigma_\lambda \in \{1.0, 1.5, \dots, 10.0\}$ and ϵ to 1×10^{-8} . All experiments were conducted on a single NVIDIA GeForce RTX 4090 GPU.

4.2. Main Results

Table 1 summarizes multivariate forecasting performance, comparing the TVA-enhanced MLP with state-of-the-art Transformer- and TCN-based baselines. Incorporating TVA into the MLP backbone consistently improved forecasting accuracy across datasets and prediction horizons, reducing the average MSE by approximately 1% to over 15%. On the Solar dataset, the MLP with TVA achieved an MSE of

Table 2. Comparison of different normalization techniques for diverse models, with and without the TVA method. Results are averaged over all prediction lengths. +TVA indicates whether TVA is applied. The x-mark denotes the normalization method alone, while the check-mark indicates its combination with TVA. The lower MSE values are highlighted in **bold**.

Models	iTransformer						DLinear			MLP			
Norm. +TVA	RevIN x ✓	DishTS x ✓	SAN x ✓	RevIN x ✓	DishTS x ✓	SAN x ✓	RevIN x ✓	DishTS x ✓	SAN x ✓	RevIN x ✓	DishTS x ✓	SAN x ✓	
ETTh1	0.449 0.443	0.484 0.470	0.657 0.529	0.446 0.446	0.477 0.468	0.651 0.511	0.448 0.438	0.484 0.487	0.544 0.469				
ETTh2	0.379 0.378	0.816 0.812	0.406 0.404	0.374 0.377	0.636 0.596	0.403 0.403	0.374 0.371	0.548 0.548	0.400 0.398				
ETTh1	0.413 0.410	0.421 0.415	0.493 0.463	0.407 0.399	0.411 0.428	0.466 0.415	0.393 0.389	0.420 0.416	0.437 0.405				
ETTh2	0.293 0.289	0.409 0.390	0.291 0.291	0.289 0.283	0.357 0.336	0.291 0.291	0.290 0.278	0.310 0.312	0.303 0.291				
Electricity	0.190 0.187	0.214 0.247	0.185 0.196	0.221 0.207	0.276 0.276	0.223 0.217	0.218 0.195	0.238 0.270	0.202 0.189				
Solar	0.298 0.276	0.275 0.256	0.266 0.256	0.352 0.269	0.330 0.281	0.303 0.251	0.277 0.234	0.279 0.264	0.256 0.250				
Traffic	0.435 0.445	0.448 0.474	0.489 0.480	0.630 0.503	0.671 0.585	0.614 0.550	0.508 0.452	0.551 0.535	0.508 0.493				
Weather	0.275 0.270	0.365 0.363	0.275 0.270	0.273 0.270	0.299 0.335	0.265 0.269	0.273 0.260	0.313 0.334	0.267 0.265				

Table 3. Comparison of different STD methods with and without TVA. Results are averaged over all prediction lengths. The x-mark denotes the STD method alone, while the check-mark indicates its combination with TVA. The lower MSE values are highlighted in **bold**.

Models	CLinear	LDLinear	DLinear	Linear
+TVA	x ✓	x ✓	x ✓	x ✓
ETTh1	0.444 0.442	0.435 0.433	0.446 0.446	0.447 0.447
ETTh2	0.392 0.384	0.375 0.369	0.374 0.377	0.375 0.382
ETTh1	0.394 0.389	0.392 0.388	0.407 0.399	0.407 0.400
ETTh2	0.295 0.272	0.287 0.280	0.289 0.283	0.298 0.283
ECL	0.185 0.177	0.191 0.189	0.221 0.207	0.219 0.206
Solar	0.312 0.249	0.288 0.263	0.352 0.269	0.343 0.263
Traffic	0.483 0.477	0.447 0.449	0.630 0.503	0.629 0.503
WTH	0.258 0.252	0.262 0.261	0.273 0.270	0.272 0.270

0.234, corresponding to an improvement of approximately 16% over the vanilla MLP. These results indicate that TVA functions as an effective parameter-level regularizer, mitigating overfitting through parameter dynamics control without architectural modifications or additional trainable parameters. Despite its structural simplicity, the TVA-enhanced MLP is frequently competitive with, and sometimes superior to, higher-capacity models. It achieved the lowest MSE on the ETT series, Solar, and Weather benchmarks among the considered baselines. On the remaining datasets, the method consistently ranked first or second in terms of both MSE and MAE, demonstrating stable forecasting behavior under varying sampling frequencies. Overall, these findings highlight that combining a minimal backbone with TVA yields strong predictive accuracy and robustness while maintaining substantially lower computational complexity than modern large-scale forecasting architectures.

4.3. Analysis

Combining TVA with instance normalization methods for diverse models. As shown in Table 2, TVA consistently improved forecasting accuracy across diverse backbones, including iTransformer, DLinear, and MLP, under

Table 4. Efficiency comparison on the Electricity dataset with look-back window $L = 96$ and prediction length $T = 720$.

Model	Parameters	Time (ms)	MSE ↓
Linear	69.8K	8.0	0.266
Linear + TVA	69.8K	9.0	0.249
DLinear	139.7K	8.2	0.270
DLinear + TVA	139.7K	9.3	0.251
MLP	0.94M	7.8	0.274
MLP + TVA	0.94M	9.2	0.253
iTransformer	5.15M	43.4	0.275
iTransformer + TVA	5.15M	51.9	0.244

multiple normalization schemes. For example, on the Solar dataset, adding TVA yielded an average 10.8% reduction in MSE across the tested configurations compared to using the corresponding normalization method alone. These results support two conclusions. First, TVA is architecture-agnostic. The observed gains are not tied to a specific architecture or normalization pipeline. Second, TVA complements data-level normalization. Whereas instance normalization primarily reduces nuisance variation and correlation in the input representation, TVA acts at the parameter level by regulating the transition-variance from optimization.

Compatibility of TVA with STD techniques. Table 3 compares several STD techniques with and without TVA, and also includes a vanilla Linear model as a reference baseline. As shown in Table 3, adding TVA to the Linear model often matched or even outperformed more complex STD-based architectures that do not incorporate parameter dynamics control. For instance, on the Solar dataset, Linear+TVA attained an average MSE of 0.263, improving upon the standard versions of CLinear (0.312), LDLinear (0.288), and DLinear (0.352). TVA also remained compatible with STD-based pipelines, since integrating TVA into CLinear, LDLinear, or DLinear typically yields additional gains relative to the corresponding STD method alone. Overall, STD and TVA address complementary failure modes.

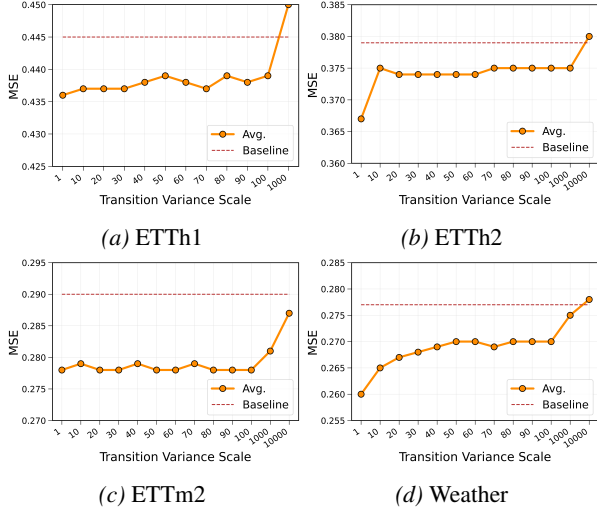


Figure 2. Performance of the MLP model with different transition-variance scales. Results are averaged over all prediction lengths. Dotted lines represent the baseline performance without TVA.

STD stabilizes the input by separating predictable components, while TVA mitigates optimization-induced overfitting by regulating the transition-variance during training. Combining stabilized representations with controlled parameter dynamics, therefore, improves generalization. This synergy is reflected in the generally improved performance observed when TVA is enabled (check-marks) compared to the corresponding STD-only configurations (x-marks).

Efficiency. TVA is computationally lightweight, as it introduces only a scalar gating operation that modulates the optimizer step size, without requiring any architectural modifications or additional trainable parameters. As reported in Table 4, TVA incurs only a modest increase in per-batch runtime, ranging from 1.0 ms to 1.4 ms for lightweight models and 8.5 ms for iTransformer. Despite this minimal overhead, TVA reduced MSE across all backbones. For example, Linear and DLinear models achieved noticeable error reductions, while the TVA-enhanced MLP improved MSE from 0.274 to 0.253. In addition, the iTransformer augmented with TVA reduced MSE from 0.275 to 0.244, demonstrating that TVA remains effective even when applied to high-capacity models. Overall, these results indicate a favorable trade-off between accuracy and efficiency. TVA operates at the optimization level without introducing architectural modifications or additional trainable parameters, making it a practical and scalable regularization strategy for MTSF.

Sensitivity Analysis. The hyperparameter σ_λ sets the target scale of the transition-variance and therefore controls the regularization strength in TVA. To assess sensitivity, we varied σ_λ over a broad range. Figure 2 reports MSE on ETTh1, ETTh2, ETTm2, and Weather, averaged over all prediction lengths $T \in \{96, 192, 336, 720\}$ with a fixed look-back win-

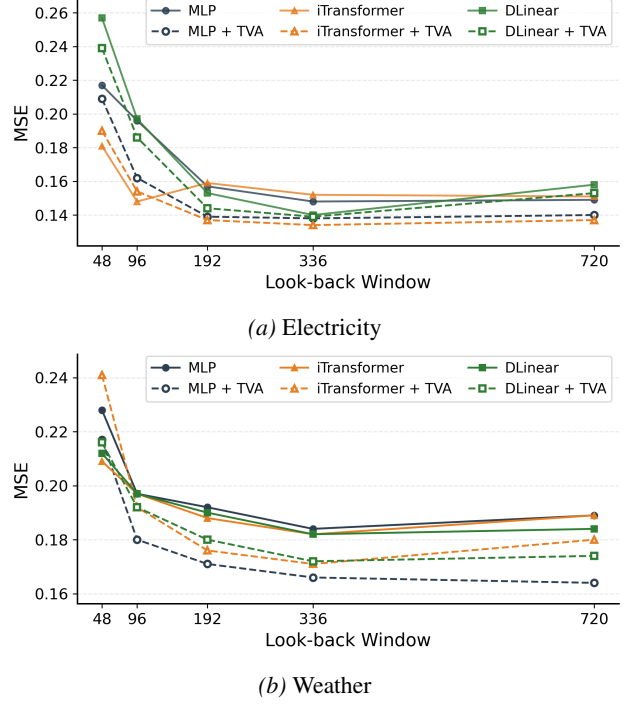


Figure 3. Performance of the MLP model under different look-back windows $L \in \{48, 96, 192, 336, 720\}$ on Electricity and Weather datasets. The prediction length T is fixed to 96.

dow $L = 96$. TVA exhibited stable performance across a wide spectrum of target scales. On ETTh1, MSE remained consistently below the baseline for σ_λ values up to 100, with noticeable degradation only at an extreme value of 1000. On ETTm2, performance remained stable up to 1000 and continues to outperform the baseline even at 10000. Although ETTh2 and Weather showed comparatively higher sensitivity, TVA remained robust over a practically relevant range of σ_λ . These findings suggest that TVA can stabilize parameter dynamics without requiring exhaustive hyperparameter tuning. In practice, the method is robust to the choice of σ_λ , substantially reducing the burden of hyperparameter search and improving applicability across diverse scenarios.

Performance on Various Look-back Window. Longer look-back windows can introduce redundant history, amplify noise, and increase exposure to non-stationary shifts, which often makes optimization harder and degrades generalization (Zeng et al., 2023; Liu et al., 2023). We therefore evaluated TVA under varying look-back windows $L \in \{48, 96, 192, 336, 720\}$ while fixing the prediction length to $T = 96$. As shown in Figure 3, the baseline MLP exhibited the typical trend that performance deteriorated as L became large, whereas the TVA-enhanced MLP remained noticeably more stable. TVA generally improved accuracy across the tested L , with the gains becoming more apparent in the long-context regime, where redundant or weakly informative history most strongly harms the baseline. This

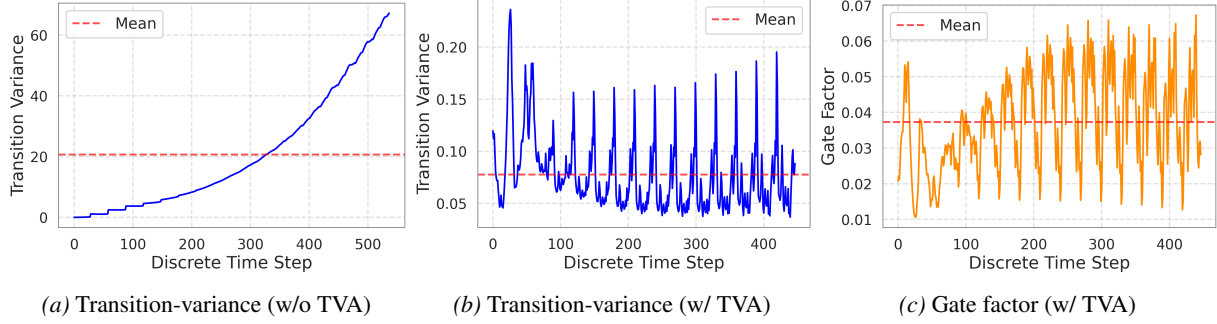


Figure 4. Transition-variance dynamics with and without TVA, together with the corresponding adaptive gate behavior on the ETTh1 dataset. The look-back window L is fixed to 96, and the prediction length T is fixed to 720. Additional results on various datasets are presented in Appendix C.1.

pattern is consistent with TVA’s role as a parameter-level regularizer. Longer look-back windows induce stronger dependence among sliding-window mini-batches and more persistent gradient noise, which can inflate the transition-variance and yield overly correlated parameter updates. By damping such unstable updates, it mitigates the performance degradation that typically arises when overly long context is used.

Transition-Variance Dynamics. To examine whether TVA actively regulates transition-variance dynamics during training, we visualized the transition-variance trajectories together with the adaptive gate behavior. As shown in Figure 4(a), without TVA, the transition-variance progressively increased throughout training, indicating unstable parameter transitions and amplified update dynamics. In contrast, Figure 4(b) demonstrates that TVA kept the transition-variance bounded, with fluctuations concentrated around a stable regime. This bounded behavior is consistent with the theoretical motivation in Section 3, where controlling transition-variance mitigates optimization-induced dependence and prevents the accumulation of excessive temporal dependencies in parameter dynamics. Figure 4(c) further illustrates that the adaptive gate factor α_k varied dynamically rather than remaining fixed throughout training. Several pronounced decreases in α_k were observed, which were consistent with periods where temporary transition-variance increases appeared in Figure 4(b). These observations suggest that TVA constrains transition-variance growth through adaptive step-size modulation, rather than by uniformly reducing update magnitudes. Additional results on various datasets are provided in Appendix C.1.

5. Conclusion

We studied multivariate time-series forecasting from the parameter-decorrelation perspective, complementing prior work that primarily focuses on data decorrelation. By modeling stochastic optimization as a parameter-space Markov process, we derived the transition-variance signal that cap-

tures optimization-induced dependence in the parameter dynamics, and showed that inflation of this signal can degrade generalization. To mitigate this effect, we proposed TVA, a architecture-agnostic method that controls the transition-variance during training via a smooth step-size gate. Extensive experiments on real-world benchmarks demonstrate that TVA consistently improves forecasting accuracy across diverse architectures with negligible computational overhead, while integrating seamlessly with existing approaches. The consistent gains observed when TVA is combined with the data decorrelation techniques further highlight the complementary value of regulating optimization-induced dependence. Overall, our findings emphasize the importance of controlling parameter dynamics for improved generalization in multivariate time-series forecasting.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Alquier, P. User-friendly Introduction to PAC-Bayes Bounds. *arXiv preprint arXiv:2110.11216*, 2021.
- Ansari, A. F., Heng, A., Lim, A., and Soh, H. Neural Continuous-Discrete State Space Models for Irregularly-

- Sampled Time Series. In *Proceedings of International Conference on Machine Learning (ICML)*, 2023.
- Barlin, J. N., Zhou, Q., Clair, C. M. S., Iasonos, A., Soslow, R. A., Alektiar, K. M., Hensley, M. L., Leita Jr, M. M., Barakat, R. R., and Abu-Rustum, N. R. Classification and regression tree (CART) analysis of endometrial carcinoma: seeing the forest for the trees. *Gynecologic oncology*, pp. 452–456, 2013.
- Cai, W., Liang, Y., Liu, X., Feng, J., and Wu, Y. MSGNet: Learning Multi-Scale Inter-series Correlations for Multivariate Time Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 11141–11149, 2024.
- Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System. In *Proceedings of International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 785–794, 2016.
- Cortes, C. and Vapnik, V. Support-vector networks. *Machine learning*, pp. 273–297, 1995.
- Das, A., Kong, W., Leach, A., Mathur, S. K., Sen, R., and Yu, R. Long-term Forecasting with TiDE: Time-series Dense Encoder. *Transactions on Machine Learning Research*, 2023.
- Fan, W., Zheng, S., Yi, X., Cao, W., Fu, Y., Bian, J., and Liu, T. Y. DEPTS: Deep expansion learning for periodic time series forecasting. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2022.
- Fan, W., Wang, P., Wang, D., Wang, D., Zhou, Y., and Fu, Y. Dish-TS: A General Paradigm for Alleviating Distribution Shift in Time Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 7522–7529, 2023.
- Friedman, J. H. Greedy Function Approximation: A Gradient Boosting Machine. *Annals of statistics*, pp. 1189–1232, 2001.
- Han, L., Chen, X.-Y., Ye, H.-J., and Zhan, D.-C. SOFTS: Efficient multivariate time series forecasting with series-core fusion. In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Hendrycks, D. and Gimpel, K. Gaussian Error Linear Units (gelus). *arXiv:1606.08415*, 2016.
- Kim, T., Kim, J., Tae, Y., Park, C., Choi, J.-H., and Choo, J. Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2022.
- Kingma, D. P. and Ba, J. Adam: A Method for Stochastic Optimization. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2015.
- Li, Q., Tai, C., and Weinan, E. Stochastic modified equations and dynamics of stochastic gradient algorithms i: Mathematical foundations. *Journal of Machine Learning Research*, pp. 1–47, 2019.
- Li, Z., Malladi, S., and Arora, S. On the validity of modeling sgd with stochastic differential equations (sdes). In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, pp. 12712–12725, 2021.
- Lin, S., Lin, W., HU, X., Wu, W., Mo, R., and Zhong, H. CycleNet: Enhancing Time Series Forecasting through Modeling Periodic Patterns. In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, 2024a.
- Lin, S., Lin, W., Wu, W., Chen, H., and Yang, J. SparseTSF: Modeling Long-term Time Series Forecasting with 1k Parameters. In *Proceedings of International Conference on Machine Learning (ICML)*, 2024b.
- Lin, S., Lin, W., Wu, W., Chen, H., and Chen, C. P. SparseTSF: Lightweight and Robust Time Series Forecasting via Sparse Modeling. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., and Long, M. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2024.
- Liu, Z., Cheng, M., Li, Z., Huang, Z., Liu, Q., Xie, Y., and Chen, E. Adaptive Normalization for Non-stationary Time Series Forecasting: A Temporal Slice Perspective. In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, pp. 14273–14292, 2023.
- Luo, D. and Wang, X. ModernTCN: A Modern Pure Convolution Structure for General Time Series Analysis. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2024.
- Malladi, S., Lyu, K., Panigrahi, A., and Arora, S. On the SDEs and scaling rules for adaptive gradient algorithms. In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, pp. 7697–7711, 2022.
- Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2023.

- Ogasawara, E., Martinez, L. C., de Oliveira, D., Zimbrão, G., Pappa, G. L., and Mattoso, M. Adaptive Normalization: A novel data normalization approach for non-stationary time series. *International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8, 2010.
- Passalis, N., Tefas, A., Kannianen, J., Gabbouj, M., and Iosifidis, A. Deep Adaptive Input Normalization for Time Series Forecasting. *IEEE Transactions on Neural Networks and Learning Systems*, pp. 3760–3765, 2019.
- Qiu, X., Hu, J., Zhou, L., Wu, X., Du, J., Zhang, B., Guo, C., Zhou, A., Jensen, C. S., Sheng, Z., et al. TFB: Towards Comprehensive and Fair Benchmarking of Time Series Forecasting Methods. In *Proceedings of VLDB Endowment*, pp. 2363–2377, 2024.
- Quinlan, J. R. Induction of decision trees. *Machine learning*, pp. 81–106, 1986.
- Rothfuss, J., Josifoski, M., Fortuin, V., and Krause, A. Scalable PAC-Bayesian Meta-Learning via the PAC-optimal Hyper-Posterior: From Theory to Practice. *Journal of Machine Learning Research*, 24(386):1–62, 2023.
- Särkkä, S. and Solin, A. *Applied stochastic differential equations*. Cambridge University Press, 2019.
- Särkkä, S. and Svensson, L. *Bayesian Filtering and Smoothing*. Cambridge University Press, 2023.
- Seeger, M. PAC-Bayesian Generalisation Error Bounds for Gaussian Process Classification. *Journal of Machine Learning Research*, 3(Oct):233–269, 2002.
- Vapnik, V. N., Golowich, S. E., and Smola, A. Support Vector Method for Function Approximation, Regression Estimation and Signal Processing. In *Proceedings of Neural Information Processing Systems (NIPS)*, 1996.
- Wen, Q., Zhou, T., Zhang, C., Chen, W., Ma, Z., Yan, J., and Sun, L. Transformers in time series: a survey. In *Proceedings of International Joint Conference on Artificial Intelligence (IJCAI)*, 2023.
- Wu, H., Xu, J., Wang, J., and Long, M. Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting. In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., and Long, M. TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2023.
- Ye, W., Deng, S., Zou, Q., and Gui, N. Frequency adaptive normalization for non-stationary time series forecasting. In *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, pp. 31350–31379, 2024.
- Yu, G., Zou, J., Hu, X., Aviles-Rivero, A. I., Qin, J., and Wang, S. Revitalizing Multivariate Time Series Forecasting: Learnable Decomposition with Inter-Series Dependencies and Intra-Series Variations Modeling. In *Proceedings of International Conference on Machine Learning (ICML)*, 2024.
- Zeng, A., Chen, M., Zhang, L., and Xu, Q. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 11121–11128, 2023.
- Zhang, Y. and Yan, J. Crossformer: Transformer Utilizing Cross-Dimension Dependency for Multivariate Time Series Forecasting. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2023.
- Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., and Zhang, W. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 11106–11115, 2021.
- Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., and Jin, R. FEDformer: Frequency Enhanced Decomposed Transformer for Long-term Series Forecasting. In *Proceedings of International Conference on Machine Learning (ICML)*, 2022.

A. More Details of TVA

For completeness, we provide proof and additional justification for the proposed method.

A.1. Proof of Lemma 3.1

Proof of Lemma 3.1. We prove the PAC-Bayes inequality and then show that a variance-diverging Gaussian posterior makes the resulting complexity term diverge.

A λ -form PAC-Bayes inequality (Alquier, 2021). Fix any $\lambda > 0$. For a fixed $\theta \in \mathbb{R}^d$, define $\ell_i(\theta) \triangleq \ell(\theta, z_i) \in [0, C]$, so that $L_S(\theta) = \frac{1}{n} \sum_{i=1}^n \ell_i(\theta)$ and $R(\theta) = \mathbb{E}_{z \sim D}[\ell(\theta, z)]$. By Hoeffding's lemma applied to the bounded random variable $\ell(\theta, z) - R(\theta) \in [-R(\theta), C - R(\theta)]$ (whose range length is at most C), and using independence across $z_1, \dots, z_n$, we obtain

$$\mathbb{E}_{S \sim D^n} \left[\exp \left(\lambda (R(\theta) - L_S(\theta)) \right) \right] \leq \exp \left(\frac{\lambda^2 C^2}{8n} \right). \quad (27)$$

Integrating Eq. (27) over a prior π and applying Fubini's theorem yields

$$\mathbb{E}_{S \sim D^n} \left[\mathbb{E}_{\theta \sim \pi} \exp \left(\lambda (R(\theta) - L_S(\theta)) \right) \right] \leq \exp \left(\frac{\lambda^2 C^2}{8n} \right). \quad (28)$$

By Markov's inequality, with probability at least $1 - \delta$ over $S \sim D^n$,

$$\mathbb{E}_{\theta \sim \pi} \exp \left(\lambda (R(\theta) - L_S(\theta)) \right) \leq \exp \left(\frac{\lambda^2 C^2}{8n} \right) \frac{1}{\delta}. \quad (29)$$

On the event Eq. (29), apply the Donsker–Varadhan change-of-measure inequality: for any posterior ρ and any measurable f ,

$$\mathbb{E}_{\theta \sim \rho} [f(\theta)] \leq \text{KL}(\rho \| \pi) + \log \mathbb{E}_{\theta \sim \pi} [e^{f(\theta)}].$$

Choosing $f(\theta) = \lambda (R(\theta) - L_S(\theta))$ and using Eq. (29) gives

$$\lambda \left(\mathbb{E}_{\theta \sim \rho} [R(\theta)] - \mathbb{E}_{\theta \sim \rho} [L_S(\theta)] \right) \leq \text{KL}(\rho \| \pi) + \frac{\lambda^2 C^2}{8n} + \log \left(\frac{1}{\delta} \right).$$

Rearranging yields Eq. (12). The bound holds for any fixed $\lambda > 0$; if one wishes to choose λ data-dependently, use a discrete grid plus union bound.

Diverging posterior variance makes the complexity term diverge. Let $\pi = \mathcal{N}(0, \sigma^2 I)$ and $\rho_k = \mathcal{N}(m_k, s_k^2 I)$. The KL divergence between isotropic Gaussians admits the closed form

$$\text{KL}(\rho_k \| \pi) = \frac{1}{2} \left(\frac{\|m_k\|_2^2}{\sigma^2} + d \left(\frac{s_k^2}{\sigma^2} - 1 - \log \frac{s_k^2}{\sigma^2} \right) \right). \quad (30)$$

By assumption, $\|m_k\|_2$ is bounded, so $\|m_k\|_2^2 / \sigma^2 = O(1)$. Let $t_k \triangleq s_k^2 / \sigma^2$. Since $(s_k^2)_{k \geq 1}$ is nondecreasing and $s_k^2 \rightarrow \infty$, we have $t_k \rightarrow \infty$ and eventually $t_k \geq 1$. Define $\varphi(t) \triangleq t - 1 - \log t$, for which $\varphi'(t) = 1 - \frac{1}{t} \geq 0$ on $[1, \infty)$. Hence $\varphi(t_k)$ is eventually nondecreasing, and because $t_k \rightarrow \infty$,

$$\varphi(t_k) = t_k - 1 - \log t_k \rightarrow \infty.$$

Plugging this into Eq. (30) shows that $\text{KL}(\rho_k \| \pi) \rightarrow \infty$ and is eventually nondecreasing. Therefore, for any fixed $\lambda > 0$,

$$\mathcal{C}_k(\lambda) \triangleq \frac{\text{KL}(\rho_k \| \pi) + \log(1/\delta)}{\lambda}$$

is eventually nondecreasing and satisfies $\mathcal{C}_k(\lambda) \rightarrow \infty$ as $k \rightarrow \infty$. Since the remaining terms in Eq. (12) are finite for fixed λ and $\mathbb{E}_{\theta \sim \rho_k} [L_S(\theta)] \in [0, C]$, the right-hand side of Eq. (12) diverges along such a trajectory. Consequently, the certificate worsens for sufficiently large k and becomes vacuous. $\square$

A.2. Proof of Lemma 3.2

Proof of Lemma 3.2. Let p_k and q_k be the marginals induced by the Markov kernels $(P_i)_{i \geq 1}$ and $(Q_i)_{i \geq 1}$:

$$p_k(d\theta_k) = \int P_k(d\theta_k | \theta_{k-1}) p_{k-1}(d\theta_{k-1}), \quad q_k(d\theta_k) = \int Q_k(d\theta_k | \theta_{k-1}) q_{k-1}(d\theta_{k-1}).$$

Introduce the path measures on $(\theta_0, \dots, \theta_k)$:

$$\begin{aligned} p_{0:k}(d\theta_{0:k}) &\triangleq p_0(d\theta_0) \prod_{i=1}^k P_i(d\theta_i | \theta_{i-1}), \\ q_{0:k}(d\theta_{0:k}) &\triangleq q_0(d\theta_0) \prod_{i=1}^k Q_i(d\theta_i | \theta_{i-1}), \end{aligned}$$

so that p_k and q_k are the θ_k -marginals of $p_{0:k}$ and $q_{0:k}$, respectively.

Data processing (marginalization) reduces KL. Let Π_k be the projection map $\Pi_k(\theta_{0:k}) = \theta_k$. By the data processing inequality (i.e., contraction of KL under measurable maps),

$$\text{KL}(p_k \| q_k) = \text{KL}(\Pi_{k\#} p_{0:k} \| \Pi_{k\#} q_{0:k}) \leq \text{KL}(p_{0:k} \| q_{0:k}), \quad (31)$$

where $\Pi_{k\#}$ denotes the pushforward measure.

Chain rule for KL on Markov path measures. If $p_{0:k} \not\ll q_{0:k}$, then $\text{KL}(p_{0:k} \| q_{0:k}) = +\infty$ and the claim is trivial, so assume $p_{0:k} \ll q_{0:k}$. Using the factorization of the Radon–Nikodym derivative,

$$\frac{dp_{0:k}}{dq_{0:k}}(\theta_{0:k}) = \frac{dp_0}{dq_0}(\theta_0) \prod_{i=1}^k \frac{dP_i(\cdot | \theta_{i-1})}{dQ_i(\cdot | \theta_{i-1})}(\theta_i),$$

we obtain

$$\begin{aligned} \text{KL}(p_{0:k} \| q_{0:k}) &= \mathbb{E}_{p_{0:k}} \left[\log \frac{dp_{0:k}}{dq_{0:k}}(\theta_{0:k}) \right] \\ &= \mathbb{E}_{p_{0:k}} \left[\log \frac{dp_0}{dq_0}(\theta_0) \right] + \sum_{i=1}^k \mathbb{E}_{p_{0:k}} \left[\log \frac{dP_i(\cdot | \theta_{i-1})}{dQ_i(\cdot | \theta_{i-1})}(\theta_i) \right] \\ &= \text{KL}(p_0 \| q_0) + \sum_{i=1}^k \mathbb{E}_{\theta_{i-1} \sim p_{i-1}} [\text{KL}(P_i(\cdot | \theta_{i-1}) \| Q_i(\cdot | \theta_{i-1}))]. \end{aligned} \quad (32)$$

The last equality follows from the tower property under $p_{0:k}$ and the definition of conditional KL: conditioning on θ_{i-1} , the inner expectation becomes the KL between the conditional laws $P_i(\cdot | \theta_{i-1})$ and $Q_i(\cdot | \theta_{i-1})$.

Combine. Combining Eq. (31) with Eq. (32) yields

$$\text{KL}(p_k \| q_k) \leq \text{KL}(p_0 \| q_0) + \sum_{i=0}^{k-1} \mathbb{E}_{\theta_i \sim p_i} [\text{KL}(P_{i+1}(\cdot | \theta_i) \| Q_{i+1}(\cdot | \theta_i))],$$

which is exactly Eq. (13). The final statement follows by setting $p_0 = q_0$. $\square$

A.3. Proof of Theorem 3.3

Proof of Theorem 3.3. We prove each claim in turn.

Step 1: Marginal KL is controlled by the sum of expected transition KLs. By Lemma 3.2, for any $k \geq 1$,

$$\text{KL}(p_k \| q_k) \leq \text{KL}(p_0 \| q_0) + \sum_{i=0}^{k-1} \mathbb{E}_{\theta_i \sim p_i} [\text{KL}(P_{i+1}(\cdot | \theta_i) \| Q_{i+1}(\cdot | \theta_i))].$$

Under the assumption $p_0 = q_0$, the initial term vanishes, yielding Eq. (20).

Step 2: Closed-form transition KL for matched-mean isotropic Gaussians. Fix $i \geq 1$ and condition on θ_{i-1} . By Eq. (14)–(15), $P_i(\cdot | \theta_{i-1})$ and $Q_i(\cdot | \theta_{i-1})$ are Gaussians with identical mean $\theta_{i-1} - \eta_i g_i(\theta_{i-1})$ and isotropic covariances $\sigma_i^2 \eta_i^2 I$ and $\sigma_\lambda^2 \eta^2 I$, respectively. Therefore, by the standard Gaussian KL formula,

$$\text{KL}(P_i \| Q_i) = \frac{d}{2} \left(\frac{\sigma_i^2 \eta_i^2}{\sigma_\lambda^2 \eta^2} - 1 - \log \frac{\sigma_i^2 \eta_i^2}{\sigma_\lambda^2 \eta^2} \right).$$

With $\eta_i = \alpha_i \eta$ from Eq. (17), this becomes

$$\text{KL}(P_i \| Q_i) = \frac{d}{2} (r_i - 1 - \log r_i), \quad r_i \triangleq \frac{\sigma_i^2 \alpha_i^2}{\sigma_\lambda^2},$$

which is Eq. (21).

Step 3: The TVA gate enforces $r_i \leq 1$ (up to a negligible ε effect). Recall the gate Eq. (18):

$$\alpha_i = \min \left\{ \frac{\sigma_i + \varepsilon}{\sigma_\lambda}, \frac{\sigma_\lambda}{\sigma_i + \varepsilon} \right\}.$$

We show $\sigma_i \alpha_i \leq \sigma_\lambda$ for all $\sigma_i \geq 0$, which implies $r_i \leq 1$.

Case 1: $\sigma_i + \varepsilon \leq \sigma_\lambda$. Then $\alpha_i = \frac{\sigma_i + \varepsilon}{\sigma_\lambda}$, so

$$\sigma_i \alpha_i = \frac{\sigma_i(\sigma_i + \varepsilon)}{\sigma_\lambda} \leq \frac{(\sigma_i + \varepsilon)^2}{\sigma_\lambda} \leq \frac{\sigma_\lambda^2}{\sigma_\lambda} = \sigma_\lambda,$$

where we used $\sigma_i \leq \sigma_i + \varepsilon$ and $\sigma_i + \varepsilon \leq \sigma_\lambda$.

Case 2: $\sigma_i + \varepsilon \geq \sigma_\lambda$. Then $\alpha_i = \frac{\sigma_\lambda}{\sigma_i + \varepsilon}$, so

$$\sigma_i \alpha_i = \sigma_i \frac{\sigma_\lambda}{\sigma_i + \varepsilon} \leq \sigma_\lambda \quad \text{since} \quad \frac{\sigma_i}{\sigma_i + \varepsilon} \leq 1.$$

Thus $\sigma_i \alpha_i \leq \sigma_\lambda$ always, and therefore

$$r_i = \frac{\sigma_i^2 \alpha_i^2}{\sigma_\lambda^2} \leq 1.$$

Step 4: Uniform boundedness of $\text{KL}(P_i \| Q_i)$ in the variance-inflation regime. Assume $\sigma_i \geq \sigma_\lambda$. Then $\sigma_i + \varepsilon \geq \sigma_\lambda$ and hence $\alpha_i = \frac{\sigma_\lambda}{\sigma_i + \varepsilon}$, yielding

$$r_i = \frac{\sigma_i^2}{(\sigma_i + \varepsilon)^2} = \left(\frac{\sigma_i}{\sigma_i + \varepsilon} \right)^2.$$

The function $h(\sigma) = \frac{\sigma}{\sigma + \varepsilon}$ is increasing on $(0, \infty)$, so for $\sigma_i \geq \sigma_\lambda$,

$$r_i \in \left[\left(\frac{\sigma_\lambda}{\sigma_\lambda + \varepsilon} \right)^2, 1 \right),$$

which is Eq. (22). Define $r_\star \triangleq \left(\frac{\sigma_\lambda}{\sigma_\lambda + \varepsilon} \right)^2 \in (0, 1)$ and $\varphi(r) \triangleq r - 1 - \log r$ for $r > 0$. On $(0, 1]$ we have $\varphi'(r) = 1 - \frac{1}{r} \leq 0$, hence φ is decreasing on $(0, 1]$. Since $r_i \in [r_\star, 1)$, it follows that $\varphi(r_i) \leq \varphi(r_\star)$ and therefore

$$\text{KL}(P_i \| Q_i) = \frac{d}{2} \varphi(r_i) \leq \frac{d}{2} \varphi(r_\star) = \frac{d}{2} (r_\star - 1 - \log r_\star),$$

a finite constant independent of σ_i hence uniformly bounded over the regime $\sigma_i \geq \sigma_\lambda$. Moreover, as $\sigma_i \rightarrow \infty$ with $\varepsilon > 0$ fixed, $r_i \rightarrow 1$ and thus $\text{KL}(P_i \| Q_i) \rightarrow 0$, showing explicitly that the per-step transition KL does not blow up as σ_i becomes large.

Step 5: Conclude. Step 1 gives the marginal bound Eq. (20). Steps 2–4 show that TVA enforces $r_i \leq 1$ and yields a uniform upper bound on each transition KL term in the variance-inflation regime $\sigma_i \geq \sigma_\lambda$. Consequently, TVA prevents variance-driven explosion of the transition-KL contributions when σ_i becomes large, and therefore controls an upper bound on marginal KL drift through Eq. (20). $\square$

A.4. Overall Pseudocode of TVA Algorithm

Algorithm 1 Transition-variance alignment.

Require: Base learning rate η , Initial model parameter $\hat{\theta}_0$, Reference variance σ_λ , Dimensionality of parameter d , Time-series dataset $\{S_1, \dots, S_K\}$

$m_0 \leftarrow \hat{\theta}_0, V_0 \leftarrow \mathbf{0}, \hat{g}_0 \leftarrow 0, \epsilon = 1 \times 10^{-8}$

for $k = 1$ **to** K **do**

$g_k \leftarrow \text{Adam}(\hat{\nabla} L_{S_k}(m_k))$

$\bar{g}_k \leftarrow 1/k \sum_d (g_k + \hat{g}_{k-1})$

$\sigma_k^2 \leftarrow 1/d \sum_d (\bar{g}_k - \hat{g}_{k-1})^2$

$\alpha_k \leftarrow \exp\left(-|\log(\sigma_k + \epsilon) - \log \sigma_\lambda|\right)$

$m_k \leftarrow m_{k-1} - \alpha_k \eta g_k(m_{k-1})$

$V_k \leftarrow V_{k-1} + \sigma_k^2 (\alpha_k \eta)^2 I$

$\hat{g}_k \leftarrow (g_k + \hat{g}_{k-1})$

end for

While the theoretical development in Section 3 is presented using a generic stochastic optimization framework, our experiments employ Adam optimization in practice. Accordingly, Algorithm 1 summarizes the overall TVA procedure integrated with Adam-based updates. At each training step, TVA first computes the stochastic gradient and the corresponding Adam-preconditioned update, since the actual parameter update is determined after optimizer preconditioning. TVA then estimates the transition-variance scale from the deviation of the preconditioned update relative to its running mean using a mean-field approximation. Here, the running mean is introduced solely to construct the centered update required by the variance definition, rather than to perform EMA-style smoothing of the transition-variance itself. Based on the discrepancy between the estimation transition-variance and the target scale, TVA computes an adaptive gate factor that modulates the global step size. The resulting gated update stabilizes parameter transition dynamics while remaining fully compatible with standard adaptive optimizers such as Adam.

B. Implementation Details

B.1. Dataset Description

We evaluated the performance of TVA on eight widely used time-series forecasting datasets (Zhou et al., 2021; Wu et al., 2021) for multivariate time-series forecasting. The detailed statistics for each dataset are summarized in Table 5.

ETT series contain oil temperature and load measurements from power transformers, collected between July 2016 and July 2018. ETTh1 and ETTh2 were sampled at hourly intervals, while ETTm1 and ETTm2 were recorded at every 15-minute intervals. All four sub-datasets share seven common variables (channels).

Table 5. Statistics of benchmark datasets.

Dataset	ETTh1	ETTh2	ETTm1	ETTm2	Electricity	Solar-Energy	Traffic	Weather
Channels	7	7	7	7	321	137	862	21
Timesteps	17,420	17,420	69,680	69,680	26,304	52,560	17,544	52,696
Frequency	1 hour	1 hour	15 min	15 min	1 hour	10 min	1 hour	10 min

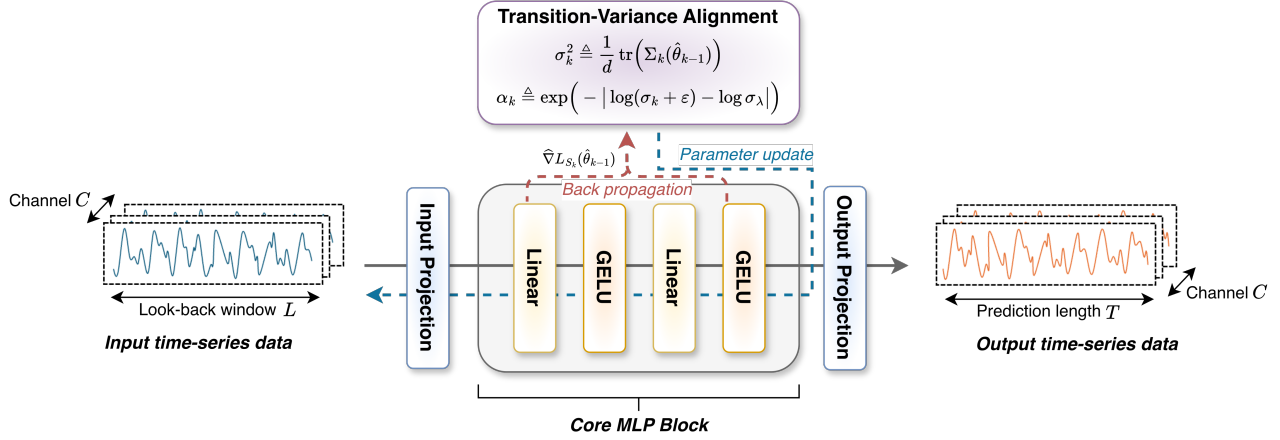


Figure 5. Overall model architecture and the integration of TVA.

Electricity (ECL) records the hourly electricity consumption of 321 customers from 2016 to 2019.

Solar-Energy (Solar) contains solar power generation data collected from 137 photovoltaic (PV) power plants in 2006, sampled at 10-minute intervals.

Traffic includes road occupancy measurements collected from 862 sensors deployed on freeways in the San Francisco Bay Area between 2015 and 2016, recorded at hourly intervals.

Weather (WTH) consists of meteorological observations collected at 10-minute intervals throughout 2020, comprising 21 indicators including temperature, humidity, and related atmospheric variables.

B.2. MLP Backbone and Integration of TVA

Figure 5 illustrates the architecture of the MLP-based backbone used in our experiments, along with the integration point of the proposed TVA algorithm. Following prior work (Lin et al., 2024a), we designed the MLP backbone as follows:

Input Projection Layer. The input tensor has shape (B, C, L) , where B is the batch size, C is the number of input variables (channels), and L is the look-back window, denoting the length of historical observations provided as input. This tensor is first permuted to (B, L, C) to better align the temporal dimension for processing. A linear transformation is then applied along the temporal axis to project the sequence of length L into a latent dimension d , enabling the model to capture temporal dependencies across channels more effectively.

Core MLP Block. The projected input is passed through two fully connected layers with GELU activation in between. This MLP block serves as the main feature extractor, transforming temporal representations into a richer latent space suitable for forecasting.

Output Projection Layer. The hidden representation is then linearly projected from the latent dimension d to the prediction length T , representing the number of future time steps to forecast. Optionally, a dropout layer may be inserted before the projection. The resulting tensor is finally permuted back to the shape (B, T, C) to match the format of the ground truth labels.

Integration of TVA. TVA operates independently of the model. It estimates parameter-wise transition-variance by comparing the current model parameters with their previous values and uses this information to interpolate between them. This mechanism adaptively suppresses inter-parameter correlation without modifying the model structure or introducing additional trainable parameters. The effect is a more stable training trajectory and improved generalization.

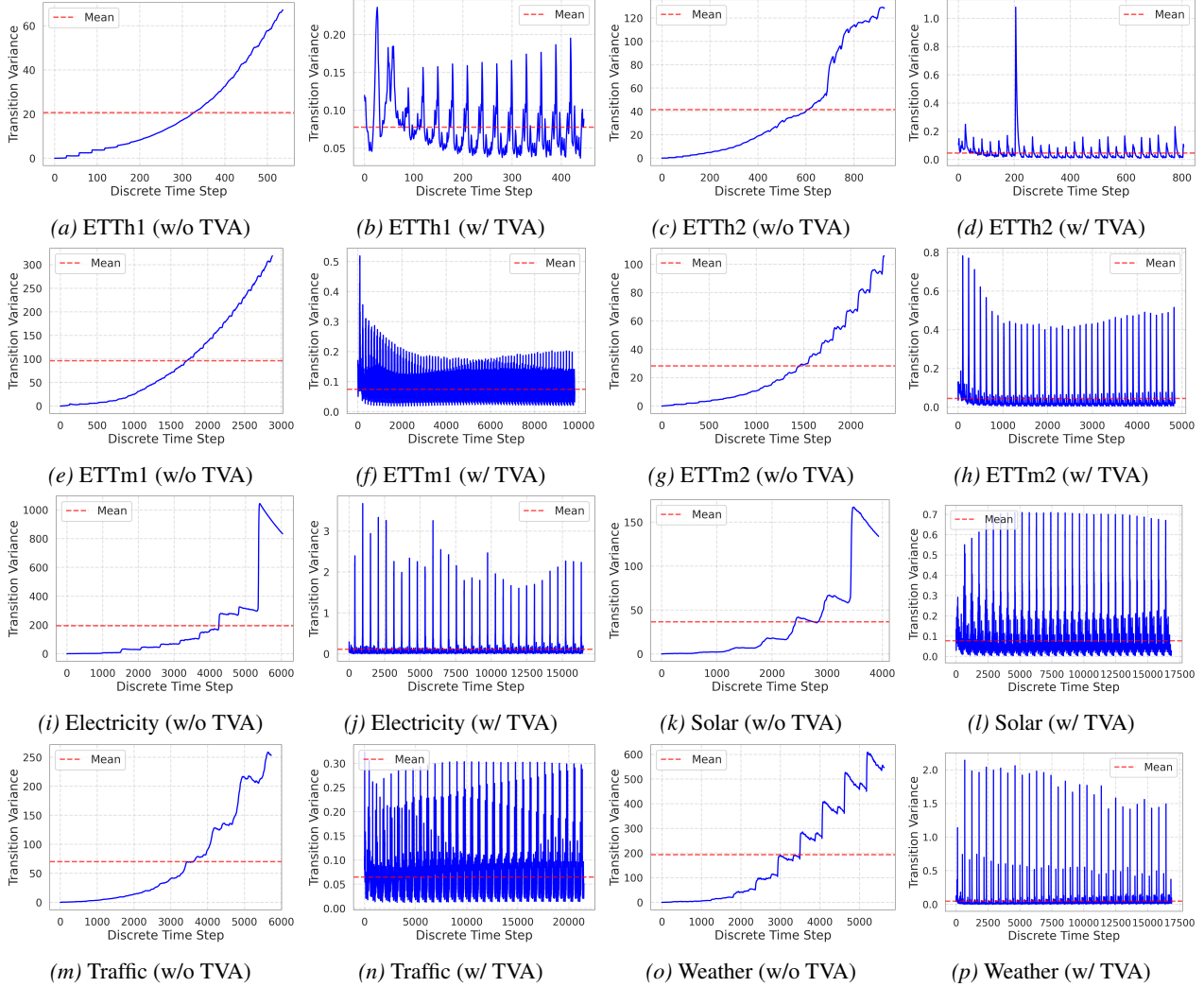


Figure 6. Transition-variance dynamics across training steps in various datasets, with and without TVA. The look-back window L is fixed to 96, and the prediction length T is fixed to 720.

C. Additional Results of TVA

C.1. Transition-variance Dynamics with and without TVA

To further support the transition-variance analysis in Section 4.3, we examined the evolution of parameter transition-variance over discrete training steps across multiple benchmark datasets. Figure 6 compares the empirical transition-variance under two configurations: (1) training without TVA (w/o TVA) and (2) training with TVA (w/ TVA). Each pair of plots corresponds to a specific dataset and illustrates how the transition-variance evolves throughout training.

In the absence of TVA, the transition-variance progressively increased across all datasets, which corresponds to unstable parameter updates. This behavior is consistent with the hypothesis discussed in the Methodology section: temporally ordered time-series samples induce correlated gradients, which amplify variance accumulation during optimization. In contrast, when TVA was applied, the transition-variance remained bounded and fluctuated around a stable mean, as indicated by the red dashed lines in Figure 6. This stabilization arises from TVA’s adaptive step-size modulation, which dynamically scales parameter updates according to the discrepancy between empirical and target transition-variance levels, thereby constraining transition-variance growth.

This effect is further corroborated by the kernel density estimation plots in Figure 7. For vanilla models, the transition-variance distributions corresponding to the initial and final training stages exhibited a significant shift, reflecting progressive drift in update uncertainty and non-stationary optimization dynamics. In contrast, when TVA was integrated, the transition-

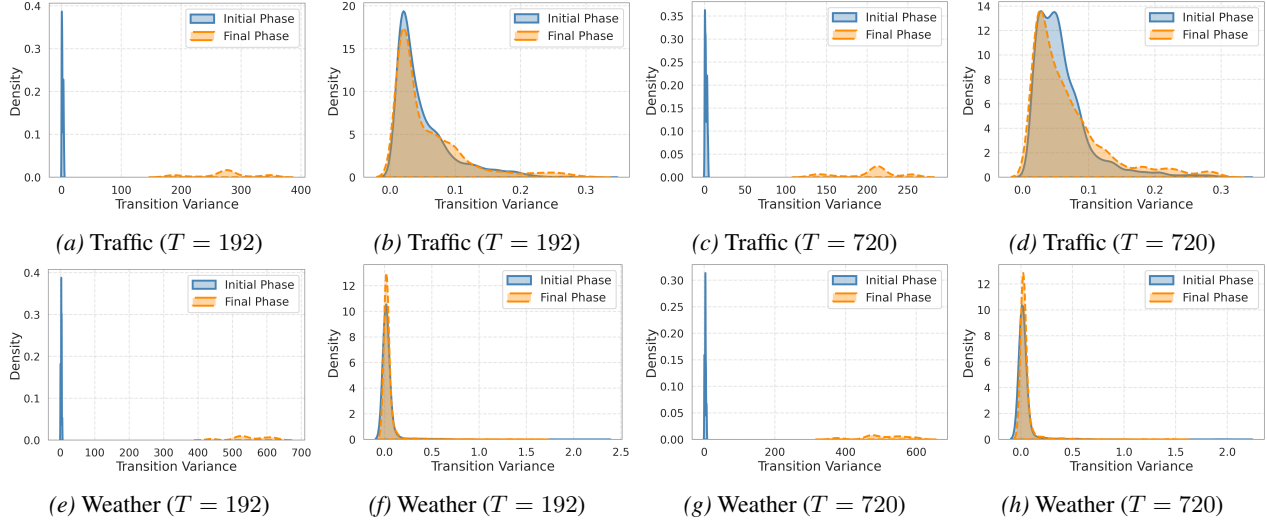


Figure 7. Kernel density estimation plots of transition-variance distributions computed from the first 20% (*Initial Phase*) and last 20% (*Final Phase*) of the sequence for the Traffic and Weather datasets. The look-back window L is fixed to 96, and the prediction length T is set to 192 or 720. Columns 1 and 3 correspond to the MLP baseline without TVA, while columns 2 and 4 show results with TVA.

variance distributions remained sharply concentrated near zero, forming narrow and high peaks, presenting predictable parameter updates. Moreover, the distributions from the initial and final training phases largely overlapped, as shown in the second and fourth columns of Figure 7. This phase-wise consistency demonstrates that cumulative drift in update uncertainty is effectively suppressed over the course of training.

By enforcing a stable transition-variance regime, TVA prevents the induced parameter distribution from progressively deviating during optimization, aligning with the theoretical motivation of controlling marginal KL divergence through bounded transition-wise variability. Overall, these results support the core idea underlying TVA: explicit regulation of transition-variance curbs excessive parameter correlation induced by temporally correlated gradients, stabilizes parameter dynamics, and ultimately improves generalization for robust multivariate time-series forecasting.

C.2. Comparison with Standard Learning-Rate Schedulers

Since TVA operates by modulating the effective step size, we examine whether its effect can be reproduced by standard learning-rate scheduling. To this end, we compare MLP+TVA with two widely used schedulers applied to the MLP backbone: cosine annealing and ReduceLROnPlateau. All experiments followed the same protocol as the main experiments. The reported results are MSE values averaged over all prediction lengths $T \in \{96, 192, 336, 720\}$.

As shown in Table 6, TVA achieved lower MSE than the standard schedulers on all datasets. While standard adaptive

Table 6. Comparison between TVA and standard learning-rate schedulers. Results are reported as MSE averaged over all prediction lengths $T \in \{96, 192, 336, 720\}$. **Bold** indicates the lowest MSE in each row, and **red values with upward arrows(↑)** denote the relative MSE increase compared to MLP+TVA.

	MLP+TVA	MLP+cosine annealing	MLP+ReduceLROnPlateau
ETTh1	0.438	0.459 (↑4.79%)	0.458 (↑4.57%)
ETTh2	0.371	0.373 (↑0.54%)	0.374 (↑0.81%)
ETTm1	0.389	0.391 (↑0.51%)	0.390 (↑0.26%)
ETTm2	0.278	0.291 (↑4.68%)	0.291 (↑4.68%)
Electricity	0.195	0.196 (↑0.51%)	0.213 (↑9.23%)
Solar	0.234	0.275 (↑17.52%)	0.262 (↑11.97%)
Traffic	0.452	0.473 (↑4.65%)	0.511 (↑13.05%)
Weather	0.260	0.268 (↑3.08%)	0.279 (↑7.31%)

Table 7. Comparison of MLP+TVA using the mean-field scale approximation and the diagonal covariance variant. Results are reported as MSE averaged over all prediction lengths $T \in \{96, 192, 336, 720\}$. ECL and WTH denote the Electricity and Weather datasets, respectively. Shaded rows indicate the default setting used in the paper.

	ETTh1	ETTh2	ETTm1	ETTm2	ECL	Solar	Traffic	WTH
MLP+TVA(mean-field scale, default)	0.438	0.371	0.389	0.278	0.195	0.234	0.452	0.260
MLP+TVA(diagonal covariance)	0.439	0.373	0.391	0.279	0.196	0.224	0.453	0.261

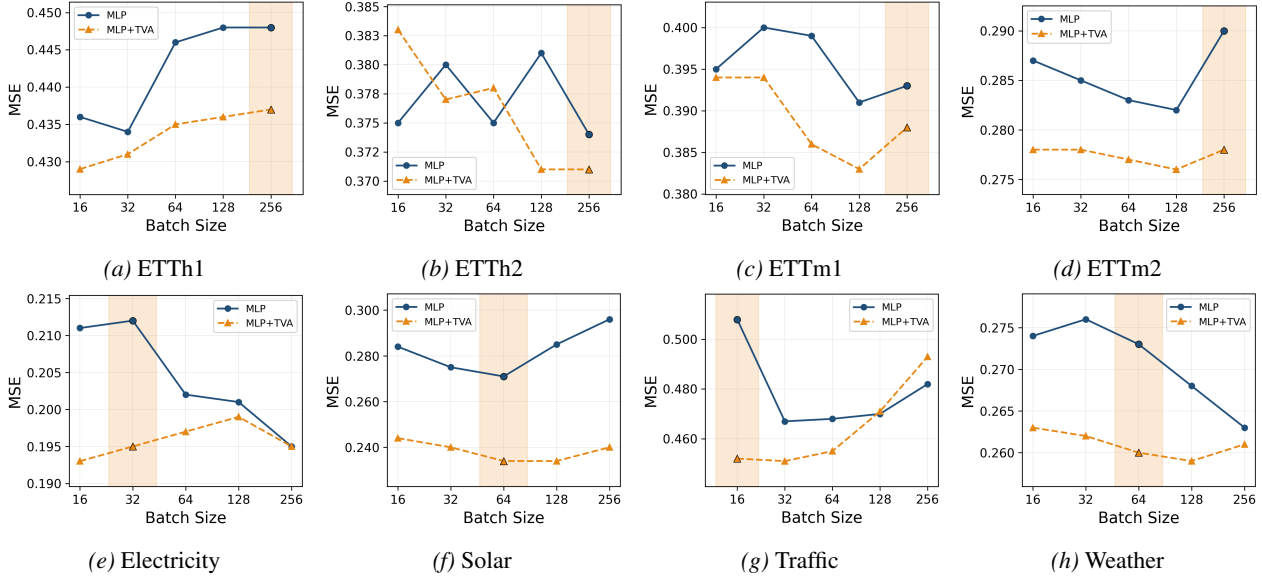


Figure 8. Batch size sensitivity analysis across various datasets. MSE is averaged over prediction lengths $T \in \{96, 192, 336, 720\}$ with a fixed look-back window $L = 96$. Shaded regions denote the default batch size configuration used throughout the paper.

learning-rate schedulers typically adjusted the step size based on external criteria, such as epoch-wise schedules or validation plateaus, TVA adjusted the effective step size at each update using the transition-variance signal σ_k^2 defined in Section 3.2. Furthermore, as described in Section 3.6, TVA modifies the optimization trajectory through the gate α_k defined in Section 3.5, thereby stabilizing the induced parameter distribution. Therefore, the advantage of TVA does not arise merely from step-size damping, but from a parameter dynamics-aware control mechanism.

C.3. Effect of Covariance Approximation

As described in Section 3.2, we used a mean-field scale approximation instead of directly estimating the full gradient-noise covariance matrix. This choice is also consistent with prior SDE-based studies (Li et al., 2019; 2021; Malladi et al., 2022), which point out that directly handling a large dense covariance matrix is impractical in high-dimensional parameter spaces.

To examine whether the mean-field scale approximation is sufficiently effective in practice, we further considered a diagonal covariance variant, which provides a richer structure than an isotropic scalar approximation. Table 7 compares the performance of MLP with TVA using the mean-field scale approximation and its diagonal covariance variant. The reported results are MSE values averaged over all prediction lengths $T \in \{96, 192, 336, 720\}$.

On the Solar dataset, using the diagonal covariance improved the MSE from 0.234 to 0.224. However, on the remaining datasets, the mean-field scale approximation achieved nearly identical or slightly better performance than the diagonal covariance variant. These results indicate that the mean-field scale approximation is sufficiently strong and efficient. Rather than being a merely convenient simplification, it represents a practical and deliberate design choice that balances predictive performance, estimation stability, and computational efficiency.

Table 8. Sensitivity analysis of MLP with and without TVA under different window overlap settings. Results are reported as MSE averaged over all prediction lengths $T \in \{96, 192, 336, 720\}$. Gain (%) denotes the relative improvement of MLP+TVA over the vanilla MLP. Shaded rows indicate the default overlap setting used in the paper.

Dataset	Overlap rate (stride)	MLP	MLP+TVA	Gain (%)
Solar	99% ($s = 1$)	0.277	0.234	+15.5
	75% ($s = 24$)	0.319	0.284	+11.0
	50% ($s = 48$)	0.438	0.425	+3.0
	0% ($s = 96$)	0.470	0.415	+11.7
Traffic	99% ($s = 1$)	0.508	0.452	+11.0
	75% ($s = 24$)	0.878	0.862	+1.8
	50% ($s = 48$)	0.859	0.838	+2.4
	0% ($s = 96$)	0.919	0.540	+41.2

C.4. Sensitivity Analysis of Batch Size

To analyze the sensitivity of TVA to batch size, we varied the batch size over $\{16, 32, 64, 128, 256\}$ and compared the performance of MLP and MLP with TVA. Figure 8 reports the MSE averaged over all prediction lengths for each dataset. Overall, MLP with TVA achieved lower MSE than the vanilla MLP across a wide range of batch sizes, showing relatively stable performance under batch size variations. In particular, on ETTh1, ETTm2, Electricity, Solar and Weather, the model with TVA consistently recorded lower errors across most batch size settings. For Solar and Weather, the vanilla MLP exhibited relatively noticeable performance fluctuations as the batch size changed, whereas MLP+TVA maintained lower MSE with only mild variations.

On some datasets, the gap between the two methods was small at certain batch sizes, and the vanilla MLP showed competitive performance in some large batch regimes, such as Traffic. Nevertheless, TVA generally exhibited more stable performance across diverse batch size settings and achieved lower MSE than the vanilla MLP on most datasets. This indicates that TVA is not a method that depends on a specific batch size; rather, it maintains stable performance by regulating the uncertainty of parameter transitions even under gradient-noise regimes that vary with batch size.

C.5. Sensitivity Analysis of Window Overlap

To analyze the sensitivity of TVA to window overlap, we controlled the overlap rate by varying the stride in sliding-window sampling. Specifically, we used strides $s \in \{1, 24, 48, 96\}$, corresponding to approximately 99%, 75%, 50%, and 0% overlap, respectively, and compared the performance of MLP and MLP with TVA on the Solar and Traffic datasets. Here, the 99% overlap setting corresponds to stride $s = 1$, which is the default setting used in the paper. Table 8 reports the MSE averaged over all prediction lengths for each overlap setting together with the relative improvement achieved by TVA.

As shown in Table 8, MLP with TVA consistently achieved lower MSE than the vanilla MLP across all overlap settings on both Solar and Traffic. Relative improvements ranged from +3.0% to +15.5% on Solar and from +1.8% to +41.2% on Traffic, demonstrating that TVA remains effective under diverse overlap regimes. These results indicate that the effectiveness of TVA does not depend on a specific overlap configuration. Rather, TVA maintains robust performance under varying temporal redundancy and gradient-noise regimes induced by different window overlap rates.

C.6. Visualization of Prediction Results

To qualitatively evaluate the impact of TVA, we visualized forecasting results on four representative benchmark datasets: Electricity, Weather, Solar, and Traffic. Figure 9 compares the prediction performance of the MLP backbone with and without TVA for prediction lengths $T \in \{336, 720\}$ under a fixed look-back window of $L = 96$. The first and third columns correspond to the baseline models without TVA (w/o TVA), while the second and fourth columns display the results with TVA (w/ TVA) applied.

The visualizations revealed that the vanilla baseline frequently suffered from pronounced forecasting distortions, including phase shifts and amplitude damping, particularly as the prediction horizon increased. These failure modes typically arise when the model fails to maintain stable parameter dynamics under strong temporal dependencies, leading to inflation of the transition-variance and degraded generalization.

In contrast, TVA effectively mitigates these issues by adaptively regulating update uncertainty at the parameter level. By stabilizing the induced parameter distribution and controlling the transition-KL surrogate, TVA promotes accurate temporal alignment and preserves the structural scale of fluctuations across all evaluated datasets. These qualitative results demonstrate that TVA enhances forecasting reliability by aligning internal learning dynamics with the complex temporal dependencies inherent in real-world time-series data.

C.7. Comparative Analysis with the Latest MLP Models

To evaluate the competitiveness of the proposed method, we conducted a comparative analysis against recent state-of-the-art MLP-based forecasting models, namely SparseTSF (Lin et al., 2025) and SOFTS (Han et al., 2024). In this experiment, SparseTSF refers to the SparseTSF/MLP variant. Table 9 reports the MSE and MAE results, comparing a vanilla MLP augmented with TVA against these specialized architectures. The results in Table 9 showed that a vanilla MLP integrated with TVA achieved performance comparable to or even exceeded, that of recent MLP-based forecasting models, despite not relying on architectural modifications or model-specific refinements. In particular, on the Solar dataset, MLP with TVA achieved an average MSE of 0.234, substantially outperforming SparseTSF/MLP (0.288) and SOFTS (0.301). The TVA-augmented MLP also achieved the best results on ETTh2, ETTm2, and Weather. These findings suggest that parameter decorrelation during training is as critical as intricate model-specific designs for enhancing generalization performance.

C.8. Full Results of Main Table

Table 10 presents the complete multivariate forecasting results on all benchmark datasets. All experiments were conducted with a fixed look-back window of $L = 96$ and prediction lengths $T \in \{96, 192, 336, 720\}$, covering a wide range of architectures, including MLP, Transformer-based and TCN-based backbones. The results showed that TVA consistently improved forecasting accuracy over different prediction horizons and frequently achieved state-of-the-art performance relative to the corresponding baseline models in most experimental settings. This consistent improvement under varying prediction lengths and datasets supports the conclusion that TVA constitutes a robust and scalable framework, effectively stabilizing parameter dynamics to deliver reliable high-precision forecasting regardless of backbone choice or task complexity.

C.9. Full Results of Different Normalization Techniques

Table 11 presents comprehensive experimental results for combining TVA with instance normalization techniques—RevIN (Kim et al., 2022), DishTS (Fan et al., 2023), and SAN (Liu et al., 2023)—across multiple model architectures (iTransformer (Liu et al., 2024), DLinear (Zeng et al., 2023), and MLP) and datasets. All experiments were conducted with a fixed look-back window of $L = 96$. The results consistently demonstrated that the integration of TVA with instance normalization improved forecasting accuracy across most configurations. These findings provide empirical evidence for the complementary role of TVA. While instance normalization addresses input-level distribution shifts, TVA mitigates inter-parameter correlation, which is a source of generalization error not handled by input normalization, without introducing additional model complexity or requiring architectural modifications.

C.10. Full Results of STD Techniques

Table 12 presents the full comparison of different STD techniques for time-series forecasting, evaluated with and without the integration of the TVA method. Results are reported in terms of both MSE and MAE across all prediction lengths. Consistent with earlier findings, TVA enhanced the forecasting accuracy of STD-based models across multiple datasets, highlighting its compatibility with diverse decomposition strategies. Notably, even a vanilla Linear model augmented with TVA often matched or outperformed more complex STD-based architectures. This underscores the strong standalone effectiveness of the TVA method in capturing essential temporal dependencies through parameter decorrelation.

C.11. Robustness and Reproducibility

To evaluate the robustness and reproducibility of the proposed method, we conducted additional experiments using five different random seeds: {2026, 2027, 2028, 2029, 2030}. As summarized in Table 13, the MLP model with TVA exhibits remarkably low variance across all seeds. Across all datasets and forecast horizons, the standard deviations for both MSE and MAE consistently remained below 0.003 and were often exactly 0.000, indicating minimal performance fluctuation due to randomness. These results confirm the stability and reliability of the proposed approach. In particular, the low variance underscores that the observed improvements do not arise from favorable random conditions but instead stem from the

intrinsic generalizability of the method. This robustness strengthens the empirical validity of our findings and highlights the practical suitability of the proposed approach for deployment in real-world time-series forecasting scenarios.

D. Limitations

Although TVA demonstrates robust performance across a wide range of experimental settings, it currently relies on a user-specified target transition-variance. While the sensitivity analysis indicates that TVA is relatively insensitive to the precise choice of this parameter within a broad range, manual specification may limit practicality when deploying the method across heterogeneous datasets or in scenarios with limited tuning budgets. An important direction for future work is to develop an adaptive strategy for estimating or updating the target transition-variance during training. Enabling data-driven adaptation of the target variance would further improve the usability and scalability of TVA in real-world forecasting applications.

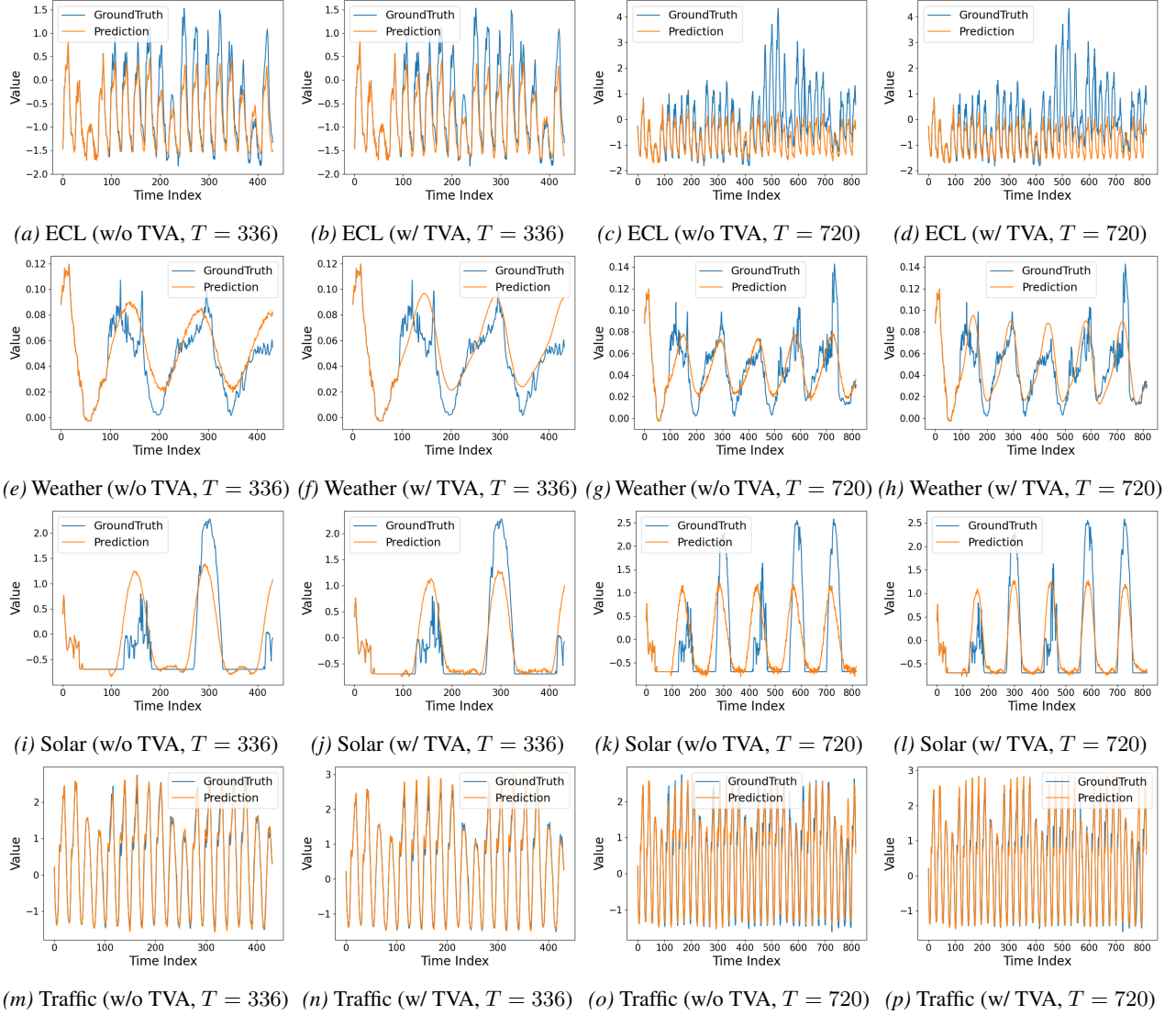


Figure 9. Visualization of forecasting results for prediction lengths $T \in \{336, 720\}$ with a fixed look-back window $L = 96$. The blue and orange lines denote the ground truth and predicted values, respectively. Columns 1 and 3 show the MLP baseline (w/o TVA), while columns 2 and 4 show the MLP with TVA (w/ TVA) results. Rows from top to bottom represent ECL, Weather, Solar, and Traffic. ECL denotes Electricity.

Table 9. Performance comparison between MLP with TVA and the latest MLP architectures. The look-back horizon was fixed at 96. Avg indicates the average performance across all prediction lengths. SparseTSF denotes SparseTSF/MLP. The lower MSE and MAE values are highlighted in **bold**.

Model		MLP		MLP+TVA		SparseTSF		SOFTS	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.386	0.398	0.372	0.394	0.376	0.390	0.380	0.403
	192	0.438	0.427	0.428	0.426	0.425	0.420	0.429	0.431
	336	0.480	0.446	0.472	0.442	0.466	0.438	0.466	0.449
	720	0.487	0.468	0.478	0.467	0.479	0.467	0.463	0.467
	Avg	0.448	0.435	0.438	0.432	0.436	0.429	0.435	0.437
ETTh2	96	0.293	0.344	0.293	0.340	0.314	0.358	0.298	0.35
	192	0.370	0.391	0.370	0.390	0.391	0.405	0.382	0.400
	336	0.416	0.428	0.409	0.423	0.428	0.436	0.423	0.434
	720	0.418	0.438	0.412	0.434	0.427	0.444	0.426	0.445
	Avg	0.374	0.400	0.371	0.397	0.390	0.411	0.382	0.407
ETTm1	96	0.339	0.363	0.317	0.357	0.344	0.376	0.341	0.377
	192	0.366	0.385	0.365	0.382	0.385	0.396	0.384	0.402
	336	0.401	0.404	0.403	0.407	0.416	0.415	0.432	0.429
	720	0.465	0.444	0.470	0.446	0.478	0.448	0.490	0.468
	Avg	0.393	0.399	0.389	0.398	0.406	0.409	0.411	0.419
ETTm2	96	0.191	0.276	0.175	0.259	0.187	0.268	0.192	0.278
	192	0.254	0.314	0.242	0.303	0.250	0.308	0.251	0.312
	336	0.313	0.350	0.299	0.338	0.308	0.345	0.333	0.366
	720	0.402	0.400	0.397	0.395	0.406	0.401	0.434	0.419
	Avg	0.290	0.335	0.278	0.324	0.288	0.330	0.302	0.344
ECL	96	0.196	0.276	0.162	0.255	0.207	0.282	0.152	0.244
	192	0.183	0.267	0.172	0.263	0.216	0.299	0.166	0.257
	336	0.218	0.298	0.192	0.284	0.226	0.308	0.186	0.279
	720	0.274	0.354	0.253	0.337	0.266	0.342	0.245	0.332
	Avg	0.218	0.299	0.195	0.285	0.229	0.308	0.187	0.278
Solar	96	0.247	0.332	0.217	0.298	0.265	0.342	0.268	0.292
	192	0.270	0.339	0.237	0.318	0.290	0.358	0.302	0.316
	336	0.282	0.357	0.238	0.314	0.292	0.356	0.318	0.327
	720	0.308	0.379	0.244	0.308	0.306	0.361	0.315	0.321
	Avg	0.277	0.352	0.234	0.310	0.288	0.354	0.301	0.314
Traffic	96	0.449	0.289	0.418	0.287	0.596	0.359	0.391	0.267
	192	0.511	0.331	0.441	0.294	0.570	0.344	0.417	0.278
	336	0.521	0.334	0.458	0.301	0.585	0.358	0.433	0.286
	720	0.551	0.354	0.491	0.320	0.645	0.375	0.463	0.302
	Avg	0.508	0.327	0.452	0.300	0.599	0.359	0.426	0.283
WTH	96	0.197	0.236	0.182	0.220	0.206	0.266	0.204	0.255
	192	0.243	0.272	0.225	0.258	0.264	0.306	0.254	0.289
	336	0.292	0.307	0.280	0.297	0.314	0.340	0.309	0.328
	720	0.360	0.351	0.354	0.346	0.381	0.381	0.379	0.373
	Avg	0.273	0.291	0.260	0.280	0.291	0.323	0.286	0.311

Table 10. Multivariate time-series forecasting results with prediction length $T \in \{96, 192, 336, 720\}$. Avg indicates the average performance across all prediction lengths. The best and second-best results are highlighted in **bold** and underlined, respectively.

Model		MLP		MLP+TVA		iTransformer		PatchTST		MSGNet		Crossformer		TimesNet	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.386	0.398	0.372	0.394	0.386	0.402	<u>0.383</u>	0.399	0.392	0.414	0.432	0.450	0.408	0.420
	192	0.438	0.427	0.428	0.426	0.441	0.434	<u>0.437</u>	0.429	0.449	0.453	0.522	0.515	0.448	0.445
	336	<u>0.480</u>	<u>0.446</u>	0.472	0.442	<u>0.480</u>	0.451	0.487	0.452	0.536	0.503	0.718	0.646	0.499	0.472
	720	<u>0.487</u>	<u>0.468</u>	0.478	0.467	<u>0.489</u>	0.479	0.499	0.476	0.497	0.490	0.646	0.604	0.515	0.496
	Avg	<u>0.448</u>	<u>0.435</u>	0.438	0.432	0.449	0.442	0.451	0.439	0.469	0.465	0.580	0.554	0.468	0.458
ETTh2	96	0.293	<u>0.344</u>	0.293	0.340	<u>0.301</u>	0.350	0.318	0.359	0.311	0.363	0.644	0.550	0.326	0.370
	192	0.370	<u>0.391</u>	0.370	0.390	<u>0.380</u>	0.399	0.391	0.404	0.415	0.425	0.629	0.577	0.408	0.421
	336	<u>0.416</u>	<u>0.428</u>	0.409	0.423	0.417	0.431	0.433	0.436	0.418	0.436	0.801	0.653	0.441	0.445
	720	<u>0.418</u>	<u>0.438</u>	0.412	0.434	0.419	0.442	0.437	0.451	0.427	0.446	1.145	0.789	0.448	0.458
	Avg	<u>0.374</u>	<u>0.400</u>	0.371	0.397	0.379	0.406	0.395	0.412	0.393	0.418	0.805	0.642	0.406	0.423
ETTm1	96	0.339	0.363	0.317	0.357	0.351	0.379	<u>0.328</u>	<u>0.362</u>	0.333	0.374	0.444	0.472	0.330	0.377
	192	<u>0.366</u>	<u>0.385</u>	0.365	0.382	0.387	0.396	0.370	0.384	0.399	0.407	0.462	0.480	0.402	0.408
	336	0.401	0.404	<u>0.403</u>	0.407	0.425	0.420	<u>0.403</u>	<u>0.405</u>	0.462	0.442	0.524	0.530	0.422	0.422
	720	<u>0.465</u>	<u>0.444</u>	0.470	0.446	0.490	0.456	0.460	0.438	0.508	0.464	0.576	0.562	0.484	0.457
	Avg	0.393	0.399	0.389	<u>0.398</u>	0.413	0.413	<u>0.390</u>	0.397	0.425	0.422	0.502	0.511	0.409	0.416
ETTm2	96	0.191	0.276	0.175	0.259	0.186	0.271	<u>0.182</u>	0.268	0.183	<u>0.265</u>	0.328	0.408	0.191	0.271
	192	0.254	0.314	0.242	0.303	0.256	0.317	<u>0.248</u>	0.310	<u>0.248</u>	0.305	0.324	0.400	0.259	0.313
	336	<u>0.313</u>	<u>0.350</u>	0.299	0.338	0.317	0.354	<u>0.313</u>	0.351	0.315	0.347	0.499	0.497	0.320	0.348
	720	<u>0.402</u>	<u>0.400</u>	0.397	0.395	0.414	0.407	0.410	0.407	0.417	0.406	0.777	0.642	0.414	0.402
	Avg	0.290	0.335	0.278	0.324	0.293	0.337	<u>0.288</u>	0.334	0.291	<u>0.331</u>	0.482	0.487	0.296	0.333
Electricity	96	0.196	0.276	<u>0.162</u>	<u>0.255</u>	0.148	0.241	0.238	0.328	0.321	0.400	0.226	0.333	0.198	0.301
	192	0.183	0.267	<u>0.172</u>	<u>0.263</u>	0.162	0.255	0.234	0.325	0.299	0.381	0.230	0.331	0.213	0.312
	336	0.218	0.298	<u>0.192</u>	<u>0.284</u>	0.176	0.270	0.255	0.345	0.302	0.383	0.308	0.390	0.223	0.322
	720	<u>0.274</u>	<u>0.354</u>	0.253	0.337	0.275	<u>0.354</u>	0.309	0.387	0.426	0.476	0.544	0.536	0.320	0.390
	Avg	0.218	0.299	<u>0.195</u>	<u>0.285</u>	0.190	0.280	0.259	0.346	0.337	0.410	0.327	0.398	0.238	0.331
Solar	96	0.247	0.332	0.217	0.298	0.272	0.300	0.304	0.352	0.860	0.673	<u>0.234</u>	0.281	0.274	0.296
	192	<u>0.270</u>	0.339	0.237	0.318	0.294	<u>0.313</u>	0.320	0.347	0.741	0.576	0.281	0.322	0.304	0.308
	336	<u>0.282</u>	0.357	0.238	0.314	0.328	0.332	0.359	0.356	0.495	0.445	0.283	0.316	0.338	0.324
	720	0.308	0.379	0.244	0.308	<u>0.297</u>	0.314	0.414	0.419	0.403	0.390	0.290	<u>0.323</u>	0.350	0.335
	Avg	0.277	0.352	0.234	0.310	0.298	0.315	0.349	0.369	0.625	0.521	<u>0.272</u>	<u>0.311</u>	0.316	0.316
Traffic	96	0.449	0.289	<u>0.418</u>	<u>0.287</u>	0.407	0.276	0.543	0.352	0.617	0.329	0.554	0.312	0.600	0.323
	192	0.511	0.331	<u>0.441</u>	<u>0.294</u>	0.425	0.283	0.530	0.342	0.655	0.349	0.581	0.333	0.630	0.336
	336	0.521	0.334	<u>0.458</u>	<u>0.301</u>	0.440	0.290	0.533	0.340	0.663	0.360	0.602	0.325	0.655	0.349
	720	0.551	0.354	<u>0.491</u>	<u>0.320</u>	0.469	0.307	0.608	0.370	0.690	0.383	0.656	0.347	0.695	0.367
	Avg	0.508	0.327	<u>0.452</u>	<u>0.300</u>	0.435	0.289	0.554	0.351	0.656	0.355	0.598	0.329	0.645	0.344
Weather	96	0.197	0.236	<u>0.182</u>	0.220	0.197	0.235	0.211	0.246	0.183	0.229	0.226	0.302	0.177	0.223
	192	0.243	0.272	0.225	0.258	0.242	0.272	0.252	0.281	<u>0.229</u>	<u>0.268</u>	0.363	0.435	0.237	0.271
	336	0.292	<u>0.307</u>	0.280	0.297	0.294	0.309	0.295	0.308	<u>0.288</u>	0.310	0.467	0.484	<u>0.288</u>	<u>0.307</u>
	720	0.360	<u>0.351</u>	0.354	0.346	0.367	0.355	0.364	0.352	<u>0.359</u>	0.353	0.480	0.504	0.366	0.358
	Avg	0.273	0.291	0.260	0.280	0.275	0.293	0.280	0.297	<u>0.265</u>	<u>0.290</u>	0.384	0.431	0.267	<u>0.290</u>

Table 11. Comparison of different normalization techniques for diverse models, with and without the TVA method. Results are reported using MSE. +TVA indicates whether TVA is applied. The x-mark denotes the normalization method alone, while the check-mark indicates its combination with TVA. ECL and WTH denote the Electricity and Weather datasets, respectively. The lower MSE values are highlighted in **bold**.

Models		iTransformer						DLinear						MLP					
Norm. +TVA		RevIN ✗ ✓		DishTS ✗ ✓		SAN ✗ ✓		RevIN ✗ ✓		DishTS ✗ ✓		SAN ✗ ✓		RevIN ✗ ✓		DishTS ✗ ✓		SAN ✗ ✓	
ETTh1	96	0.386	0.382	0.404	0.395	0.612	0.474	0.385	0.383	0.392	0.392	0.639	0.492	0.386	0.372	0.385	0.393	0.482	0.400
	192	0.441	0.434	0.444	0.445	0.683	0.568	0.436	0.435	0.451	0.446	0.648	0.459	0.438	0.428	0.457	0.458	0.521	0.455
	336	0.480	0.473	0.536	0.500	0.654	0.551	0.480	0.477	0.511	0.499	0.667	0.548	0.480	0.472	0.515	0.503	0.558	0.496
	720	0.489	0.482	0.551	0.541	0.678	0.523	0.484	0.489	0.553	0.537	0.649	0.546	0.487	0.478	0.578	0.593	0.616	0.527
	Avg	0.449	0.443	0.484	0.470	0.657	0.529	0.446	0.446	0.477	0.468	0.651	0.511	0.448	0.438	0.484	0.487	0.544	0.469
ETTh2	96	0.301	0.300	0.806	0.869	0.321	0.322	0.289	0.297	0.380	0.328	0.318	0.315	0.293	0.293	0.296	0.329	0.309	0.307
	192	0.380	0.377	0.541	0.618	0.405	0.405	0.375	0.379	0.543	0.500	0.400	0.389	0.370	0.370	0.399	0.424	0.396	0.390
	336	0.417	0.415	0.792	0.666	0.443	0.444	0.414	0.417	0.637	0.614	0.445	0.455	0.416	0.409	0.573	0.460	0.442	0.444
	720	0.419	0.419	1.126	1.095	0.454	0.446	0.420	0.415	0.984	0.941	0.447	0.454	0.418	0.412	0.923	0.978	0.454	0.450
	Avg	0.379	0.378	0.816	0.812	0.406	0.404	0.374	0.377	0.636	0.596	0.403	0.403	0.374	0.371	0.548	0.548	0.400	0.398
ETTh1	96	0.351	0.342	0.360	0.353	0.454	0.403	0.341	0.336	0.357	0.346	0.422	0.359	0.339	0.317	0.352	0.338	0.388	0.343
	192	0.387	0.387	0.393	0.387	0.467	0.435	0.385	0.380	0.386	0.402	0.428	0.387	0.366	0.365	0.396	0.395	0.413	0.373
	336	0.425	0.424	0.427	0.422	0.494	0.467	0.418	0.409	0.417	0.441	0.482	0.420	0.401	0.403	0.432	0.432	0.438	0.421
	720	0.490	0.487	0.503	0.498	0.557	0.548	0.484	0.473	0.484	0.522	0.530	0.495	0.465	0.470	0.501	0.501	0.507	0.484
	Avg	0.413	0.410	0.421	0.415	0.493	0.463	0.407	0.399	0.411	0.428	0.466	0.415	0.393	0.389	0.420	0.416	0.437	0.405
ETTh2	96	0.186	0.183	0.233	0.219	0.189	0.189	0.186	0.177	0.187	0.197	0.189	0.189	0.191	0.175	0.201	0.198	0.192	0.189
	192	0.256	0.251	0.304	0.297	0.252	0.252	0.250	0.243	0.250	0.283	0.252	0.252	0.254	0.242	0.262	0.275	0.258	0.252
	336	0.317	0.312	0.430	0.417	0.311	0.311	0.310	0.306	0.310	0.325	0.312	0.312	0.313	0.299	0.336	0.327	0.346	0.314
	720	0.414	0.411	0.667	0.625	0.411	0.410	0.410	0.405	0.681	0.537	0.410	0.410	0.402	0.397	0.444	0.446	0.415	0.409
	Avg	0.293	0.289	0.409	0.390	0.291	0.291	0.289	0.283	0.357	0.336	0.291	0.291	0.290	0.278	0.310	0.312	0.303	0.291
ECL	96	0.148	0.154	0.144	0.178	0.153	0.151	0.197	0.183	0.239	0.220	0.195	0.185	0.196	0.162	0.208	0.211	0.172	0.161
	192	0.162	0.166	0.205	0.223	0.166	0.165	0.200	0.188	0.247	0.234	0.194	0.189	0.183	0.172	0.215	0.229	0.181	0.173
	336	0.176	0.183	0.224	0.251	0.178	0.229	0.218	0.206	0.270	0.249	0.209	0.249	0.218	0.192	0.235	0.293	0.210	0.190
	720	0.275	0.244	0.282	0.337	0.243	0.238	0.270	0.251	0.346	0.401	0.292	0.248	0.274	0.253	0.295	0.348	0.244	0.232
	Avg	0.190	0.187	0.214	0.247	0.185	0.196	0.221	0.207	0.276	0.276	0.223	0.217	0.218	0.195	0.238	0.270	0.202	0.189
Solar	96	0.272	0.245	0.238	0.229	0.237	0.228	0.311	0.235	0.294	0.234	0.263	0.218	0.247	0.217	0.244	0.224	0.230	0.220
	192	0.294	0.270	0.278	0.246	0.269	0.248	0.342	0.268	0.320	0.270	0.299	0.263	0.270	0.237	0.273	0.263	0.264	0.265
	336	0.328	0.289	0.283	0.260	0.281	0.271	0.373	0.285	0.346	0.305	0.324	0.254	0.282	0.238	0.285	0.264	0.270	0.260
	720	0.297	0.299	0.300	0.291	0.277	0.276	0.382	0.288	0.358	0.315	0.327	0.268	0.308	0.244	0.313	0.303	0.259	0.257
	Avg	0.298	0.276	0.275	0.256	0.266	0.256	0.352	0.269	0.330	0.281	0.303	0.251	0.277	0.234	0.279	0.264	0.256	0.250
Traffic	96	0.407	0.417	0.418	0.440	0.465	0.450	0.649	0.480	0.684	0.535	0.623	0.535	0.449	0.418	0.532	0.512	0.486	0.469
	192	0.425	0.435	0.434	0.461	0.473	0.467	0.606	0.494	0.639	0.555	0.590	0.531	0.511	0.441	0.534	0.531	0.492	0.479
	336	0.440	0.447	0.453	0.484	0.488	0.483	0.615	0.505	0.673	0.600	0.597	0.547	0.521	0.458	0.553	0.524	0.506	0.493
	720	0.469	0.480	0.489	0.510	0.530	0.520	0.649	0.535	0.689	0.648	0.647	0.586	0.551	0.491	0.584	0.574	0.549	0.529
	Avg	0.435	0.445	0.448	0.474	0.489	0.480	0.630	0.503	0.671	0.585	0.614	0.550	0.508	0.452	0.551	0.535	0.508	0.493
WTH	96	0.197	0.192	0.290	0.210	0.195	0.191	0.197	0.192	0.221	0.222	0.189	0.192	0.197	0.182	0.215	0.285	0.190	0.192
	192	0.242	0.236	0.319	0.282	0.247	0.238	0.241	0.237	0.274	0.286	0.231	0.234	0.243	0.225	0.266	0.297	0.233	0.231
	336	0.294	0.289	0.384	0.405	0.291	0.287	0.292	0.289	0.315	0.343	0.285	0.287	0.292	0.280	0.348	0.358	0.288	0.287
	720	0.367	0.362	0.466	0.554	0.367	0.361	0.363	0.361	0.384	0.488	0.356	0.364	0.360	0.354	0.421	0.397	0.358	0.351
	Avg	0.275	0.270	0.365	0.363	0.275	0.270	0.273	0.270	0.299	0.335	0.265	0.269	0.273	0.260	0.313	0.334	0.267	0.265

Table 12. Comparison of different STD techniques with and without TVA. The x-mark denotes the STD method alone, while the check-mark indicates its combination with TVA. The lower MSE and MAE values are highlighted in **bold**.

Models +TVA		CLinear				LDLinear				DLinear				Linear			
		x		✓		x		✓		x		✓		x		✓	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.378	0.389	0.394	0.404	0.379	0.395	0.377	0.394	0.385	0.393	0.383	0.395	0.385	0.393	0.383	0.395
	192	0.432	0.419	0.436	0.426	0.428	0.422	0.427	0.422	0.436	0.423	0.435	0.426	0.437	0.423	0.434	0.425
	336	0.475	0.440	0.467	0.442	0.464	0.440	0.463	0.440	0.480	0.445	0.477	0.447	0.480	0.446	0.479	0.448
	720	0.492	0.475	0.471	0.472	0.469	0.462	0.465	0.460	0.484	0.471	0.489	0.435	0.484	0.471	0.489	0.475
	Avg	0.444	0.431	0.442	0.436	0.435	0.430	0.433	0.429	0.446	0.433	0.446	0.426	0.447	0.433	0.447	0.436
ETTh2	96	0.305	0.351	0.302	0.347	0.296	0.343	0.287	0.338	0.289	0.337	0.297	0.344	0.290	0.339	0.303	0.349
	192	0.390	0.403	0.381	0.395	0.374	0.391	0.370	0.389	0.375	0.390	0.379	0.393	0.376	0.391	0.383	0.396
	336	0.439	0.438	0.425	0.430	0.414	0.425	0.408	0.422	0.414	0.425	0.417	0.426	0.415	0.426	0.421	0.429
	720	0.433	0.446	0.426	0.443	0.417	0.437	0.411	0.434	0.420	0.439	0.415	0.435	0.420	0.439	0.423	0.441
	Avg	0.392	0.409	0.384	0.404	0.375	0.399	0.369	0.396	0.374	0.398	0.377	0.400	0.375	0.399	0.382	0.404
ETTh1	96	0.322	0.359	0.331	0.366	0.331	0.362	0.325	0.361	0.341	0.362	0.336	0.360	0.341	0.362	0.335	0.361
	192	0.379	0.394	0.369	0.386	0.373	0.384	0.366	0.382	0.385	0.386	0.380	0.384	0.384	0.385	0.379	0.384
	336	0.409	0.413	0.399	0.407	0.404	0.404	0.399	0.403	0.418	0.407	0.409	0.404	0.418	0.406	0.410	0.405
	720	0.466	0.443	0.458	0.441	0.461	0.439	0.462	0.439	0.484	0.444	0.473	0.440	0.484	0.443	0.475	0.441
	Avg	0.394	0.403	0.389	0.400	0.392	0.397	0.388	0.396	0.407	0.400	0.399	0.397	0.407	0.399	0.400	0.398
ETTh2	96	0.179	0.262	0.166	0.250	0.177	0.261	0.176	0.259	0.186	0.271	0.177	0.260	0.196	0.282	0.177	0.260
	192	0.268	0.326	0.233	0.294	0.244	0.305	0.241	0.300	0.250	0.309	0.243	0.302	0.259	0.319	0.243	0.303
	336	0.326	0.361	0.293	0.333	0.315	0.351	0.302	0.339	0.310	0.347	0.306	0.342	0.318	0.355	0.308	0.344
	720	0.409	0.402	0.395	0.391	0.412	0.406	0.401	0.396	0.410	0.402	0.405	0.398	0.418	0.408	0.405	0.398
	Avg	0.295	0.338	0.272	0.317	0.287	0.331	0.280	0.323	0.289	0.332	0.283	0.325	0.298	0.341	0.283	0.326
Electricity	96	0.150	0.247	0.145	0.240	0.166	0.257	0.161	0.257	0.197	0.274	0.183	0.268	0.200	0.276	0.183	0.268
	192	0.164	0.257	0.159	0.252	0.175	0.265	0.173	0.266	0.200	0.279	0.188	0.273	0.200	0.279	0.188	0.273
	336	0.185	0.282	0.180	0.275	0.193	0.282	0.190	0.283	0.218	0.298	0.206	0.290	0.212	0.292	0.205	0.290
	720	0.242	0.330	0.222	0.311	0.232	0.313	0.230	0.315	0.270	0.347	0.251	0.327	0.266	0.341	0.249	0.325
	Avg	0.185	0.279	0.177	0.270	0.191	0.279	0.189	0.280	0.221	0.300	0.207	0.289	0.219	0.297	0.206	0.289
Solar	96	0.256	0.283	0.203	0.267	0.269	0.300	0.241	0.289	0.311	0.402	0.235	0.336	0.288	0.375	0.230	0.330
	192	0.294	0.301	0.242	0.291	0.293	0.315	0.253	0.300	0.342	0.421	0.268	0.365	0.319	0.396	0.259	0.355
	336	0.343	0.325	0.269	0.306	0.296	0.314	0.272	0.301	0.373	0.435	0.285	0.371	0.374	0.429	0.289	0.372
	720	0.356	0.333	0.284	0.321	0.295	0.310	0.285	0.309	0.382	0.436	0.288	0.376	0.391	0.441	0.273	0.358
	Avg	0.312	0.311	0.249	0.296	0.288	0.310	0.263	0.300	0.352	0.423	0.269	0.362	0.343	0.410	0.263	0.354
Traffic	96	0.483	0.323	0.468	0.321	0.423	0.276	0.434	0.290	0.649	0.385	0.480	0.333	0.648	0.385	0.480	0.332
	192	0.464	0.300	0.463	0.304	0.437	0.282	0.435	0.288	0.606	0.363	0.494	0.333	0.605	0.364	0.492	0.332
	336	0.478	0.309	0.473	0.310	0.447	0.287	0.450	0.294	0.615	0.366	0.505	0.334	0.614	0.366	0.505	0.334
	720	0.509	0.331	0.503	0.327	0.480	0.310	0.479	0.311	0.649	0.389	0.535	0.352	0.649	0.387	0.533	0.351
	Avg	0.483	0.316	0.477	0.315	0.447	0.289	0.449	0.296	0.630	0.376	0.503	0.338	0.629	0.375	0.503	0.337
Weather	96	0.171	0.216	0.170	0.217	0.192	0.230	0.187	0.226	0.197	0.235	0.192	0.230	0.195	0.234	0.192	0.230
	192	0.228	0.264	0.218	0.255	0.228	0.263	0.229	0.263	0.241	0.271	0.237	0.266	0.240	0.270	0.237	0.266
	336	0.280	0.301	0.272	0.294	0.280	0.299	0.281	0.300	0.292	0.306	0.289	0.303	0.291	0.306	0.289	0.303
	720	0.353	0.348	0.347	0.343	0.348	0.343	0.350	0.345	0.363	0.353	0.361	0.350	0.363	0.353	0.361	0.350
	Avg	0.258	0.282	0.252	0.277	0.262	0.284	0.261	0.284	0.273	0.291	0.270	0.287	0.272	0.291	0.270	0.287

Table 13. Performance of the TVA-enhanced MLP model evaluated under different random seeds {2026, 2027, 2028, 2029, 2030}. The look-back horizon was fixed at 96. Avg indicates the average performance across all prediction lengths, while Mean and STD denote the average and standard deviation, respectively.

Setup		Random Seed													
		2026		2027		2028		2029		2030		Mean		STD	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	0.372	0.394	0.372	0.393	0.373	0.392	0.373	0.394	0.372	0.392	0.372	0.393	0.000	0.001
	192	0.428	0.426	0.428	0.421	0.429	0.422	0.428	0.421	0.429	0.422	0.428	0.422	0.001	0.002
	336	0.472	0.442	0.473	0.443	0.474	0.443	0.473	0.442	0.474	0.443	0.473	0.443	0.001	0.001
	720	0.478	0.467	0.479	0.468	0.478	0.467	0.477	0.466	0.480	0.467	0.478	0.467	0.001	0.001
	Avg	0.438	0.432	0.438	0.431	0.438	0.431	0.438	0.431	0.439	0.431	0.438	0.431	0.001	0.001
ETTh2	96	0.293	0.340	0.298	0.344	0.298	0.343	0.293	0.341	0.295	0.343	0.295	0.342	0.002	0.002
	192	0.370	0.390	0.369	0.389	0.370	0.391	0.370	0.390	0.371	0.391	0.370	0.390	0.001	0.001
	336	0.409	0.423	0.409	0.422	0.409	0.423	0.408	0.422	0.409	0.423	0.409	0.423	0.000	0.000
	720	0.412	0.434	0.410	0.433	0.414	0.436	0.410	0.432	0.412	0.434	0.412	0.434	0.002	0.001
	Avg	0.371	0.397	0.371	0.397	0.373	0.398	0.370	0.396	0.372	0.398	0.371	0.397	0.001	0.001
ETThm1	96	0.317	0.357	0.317	0.358	0.318	0.359	0.318	0.358	0.318	0.358	0.318	0.358	0.001	0.001
	192	0.365	0.382	0.365	0.382	0.366	0.383	0.366	0.383	0.365	0.382	0.365	0.383	0.000	0.001
	336	0.403	0.407	0.403	0.407	0.401	0.406	0.407	0.409	0.400	0.405	0.403	0.407	0.003	0.002
	720	0.470	0.446	0.465	0.443	0.466	0.444	0.466	0.443	0.466	0.442	0.467	0.444	0.002	0.001
	Avg	0.389	0.398	0.387	0.398	0.388	0.398	0.389	0.398	0.388	0.397	0.388	0.398	0.001	0.001
ETThm2	96	0.175	0.259	0.175	0.259	0.175	0.260	0.175	0.259	0.175	0.259	0.175	0.259	0.000	0.000
	192	0.242	0.303	0.241	0.302	0.241	0.302	0.241	0.302	0.241	0.302	0.241	0.302	0.001	0.001
	336	0.299	0.338	0.299	0.338	0.301	0.340	0.299	0.338	0.300	0.339	0.300	0.339	0.001	0.001
	720	0.397	0.395	0.395	0.394	0.397	0.395	0.394	0.394	0.394	0.394	0.395	0.394	0.001	0.001
	Avg	0.278	0.324	0.278	0.323	0.278	0.324	0.277	0.323	0.278	0.324	0.278	0.324	0.000	0.000
Electricity	96	0.162	0.255	0.162	0.254	0.161	0.254	0.162	0.255	0.162	0.255	0.162	0.255	0.000	0.000
	192	0.172	0.263	0.172	0.263	0.172	0.263	0.173	0.264	0.173	0.264	0.172	0.263	0.000	0.000
	336	0.192	0.284	0.191	0.283	0.191	0.283	0.191	0.283	0.192	0.283	0.191	0.283	0.000	0.000
	720	0.253	0.337	0.256	0.339	0.252	0.336	0.254	0.337	0.251	0.335	0.253	0.337	0.002	0.002
	Avg	0.195	0.285	0.195	0.285	0.194	0.284	0.195	0.285	0.194	0.284	0.195	0.285	0.000	0.000
Solar	96	0.217	0.298	0.216	0.299	0.216	0.300	0.215	0.295	0.215	0.295	0.216	0.297	0.001	0.002
	192	0.237	0.318	0.239	0.321	0.239	0.320	0.239	0.321	0.239	0.320	0.239	0.320	0.001	0.001
	336	0.238	0.314	0.239	0.315	0.238	0.314	0.239	0.314	0.236	0.311	0.238	0.313	0.001	0.002
	720	0.244	0.308	0.244	0.307	0.244	0.306	0.245	0.308	0.244	0.305	0.244	0.307	0.000	0.001
	Avg	0.234	0.310	0.235	0.310	0.234	0.310	0.235	0.309	0.234	0.308	0.234	0.309	0.000	0.001
Traffic	96	0.418	0.287	0.417	0.286	0.419	0.286	0.418	0.286	0.417	0.285	0.418	0.286	0.001	0.001
	192	0.441	0.294	0.440	0.293	0.440	0.293	0.439	0.293	0.441	0.293	0.440	0.293	0.001	0.000
	336	0.458	0.301	0.459	0.302	0.458	0.302	0.458	0.301	0.458	0.302	0.458	0.301	0.000	0.000
	720	0.491	0.320	0.490	0.318	0.494	0.322	0.495	0.323	0.491	0.319	0.492	0.320	0.002	0.002
	Avg	0.452	0.300	0.452	0.300	0.453	0.301	0.453	0.301	0.452	0.300	0.452	0.300	0.001	0.001
Weather	96	0.182	0.220	0.181	0.220	0.182	0.220	0.181	0.220	0.180	0.219	0.181	0.220	0.001	0.001
	192	0.225	0.258	0.225	0.258	0.225	0.258	0.225	0.258	0.225	0.258	0.225	0.258	0.000	0.000
	336	0.280	0.297	0.279	0.297	0.279	0.297	0.279	0.297	0.279	0.297	0.279	0.297	0.000	0.000
	720	0.354	0.346	0.354	0.346	0.354	0.345	0.354	0.346	0.354	0.346	0.354	0.346	0.000	0.000
	Avg	0.260	0.280	0.260	0.280	0.260	0.280	0.260	0.280	0.260	0.280	0.260	0.280	0.000	0.000